

Forecast Benchmarking Report

Generated 2026-10-05 15:22

Executive Summary

Ground Truth: CHIRPS

Benchmark Coverage: Ethiopia

Forecast Window: 1-15 and 16-30

Baseline Forecast: climatology

Models Evaluated: 3

Deterministic Models: 3

Probabilistic Models: 0

Models Evaluated

Model	Category	Type	Coverage	Date Range
AIFS Single v2	Deterministic	LIVE FORECAST	AIWP	2000 to 2025
FuXi	Deterministic	LIVE FORECAST	AIWP	1998-05 to 2024-07
GraphCast	Deterministic	LIVE FORECAST	AIWP	1998-05 to 2024-07

Benchmark Inputs

Ground Truth: CHIRPS

Benchmark Coverage: Ethiopia

Forecast Window: 1-15 and 16-30

Shared Settings

Wet Day Threshold: 20

Minimum Wet Day Rainfall: 1

Wet Spell Length: 3

Forecast Benchmarking Report

Dry Spell Limit: 7

Dry Spell Search Extension: 21

Area Mask File: /Users/mesfind/opt/Research/uchicago/ETAgroMetAlx/HCWF/ETAgroMetAlx/Training/ai-weather/ai_weather/data/jjas_100mm_rainfall_mask_0p25.nc

Baseline Forecast: climatology

Available Model Outputs

Model	Windows	File
AIFS Single v2	Days 1-15	spatial_metrics_AIFS_1-15.nc
AIFS Single v2	Days 16-30	spatial_metrics_AIFS_16-30.nc
Traditional Climatology	Days 1-15	spatial_metrics_climatology_1-15.nc
Traditional Climatology	Days 16-30	spatial_metrics_climatology_16-30.nc
FuXi	Days 1-15	spatial_metrics_FuXi_1-15.nc
FuXi	Days 16-30	spatial_metrics_FuXi_16-30.nc
Traditional Climatology	Days 1-15	spatial_metrics_climatology_1-15.nc
Traditional Climatology	Days 16-30	spatial_metrics_climatology_16-30.nc
GraphCast	Days 1-15	spatial_metrics_GraphCast_1-15.nc
GraphCast	Days 16-30	spatial_metrics_GraphCast_16-30.nc
Traditional Climatology	Days 1-15	spatial_metrics_climatology_1-15.nc
Traditional Climatology	Days 16-30	spatial_metrics_climatology_16-30.nc
AIFS Single v2	Days 1-15	spatial_metrics_AIFS_1-15.nc
AIFS Single v2	Days 16-30	spatial_metrics_AIFS_16-30.nc
FuXi	Days 1-15	spatial_metrics_FuXi_1-15.nc
FuXi	Days 16-30	spatial_metrics_FuXi_16-30.nc
GraphCast	Days 1-15	spatial_metrics_GraphCast_1-15.nc
GraphCast	Days 16-30	spatial_metrics_GraphCast_16-30.nc
Traditional Climatology	Days 1-15	spatial_metrics_climatology_1-15.nc
Traditional Climatology	Days 16-30	spatial_metrics_climatology_16-30.nc

Verification Scores

Model	Lead Window	Metric	Value
AIFS Single v2	Days 16-30	False Alarm Rate fraction	0.244
AIFS Single v2	Days 16-30	Mean Absolute Error days	11.475
AIFS Single v2	Days 16-30	Miss Rate fraction	0.402
AIFS Single v2	Days 1-15	False Alarm Rate fraction	0.277
AIFS Single v2	Days 1-15	Mean Absolute Error days	5.133
AIFS Single v2	Days 1-15	Miss Rate fraction	0.406
FuXi	Days 16-30	False Alarm Rate fraction	0.163
FuXi	Days 16-30	Mean Absolute Error days	10.497
FuXi	Days 16-30	Miss Rate fraction	0.681
FuXi	Days 1-15	False Alarm Rate fraction	0.166
FuXi	Days 1-15	Mean Absolute Error days	5.123
FuXi	Days 1-15	Miss Rate fraction	0.692
GraphCast	Days 16-30	False Alarm Rate fraction	0.191
GraphCast	Days 16-30	Mean Absolute Error days	7.631
GraphCast	Days 16-30	Miss Rate fraction	0.450
GraphCast	Days 1-15	False Alarm Rate fraction	0.289
GraphCast	Days 1-15	Mean Absolute Error days	6.160
GraphCast	Days 1-15	Miss Rate fraction	0.439
Traditional Climatology	Days 16-30	False Alarm Rate fraction	0.630
Traditional Climatology	Days 16-30	Mean Absolute Error days	13.809
Traditional Climatology	Days 16-30	Miss Rate fraction	0.342
Traditional Climatology	Days 1-15	False Alarm Rate fraction	0.422
Traditional Climatology	Days 1-15	Mean Absolute Error days	14.059
Traditional Climatology	Days 1-15	Miss Rate fraction	0.602

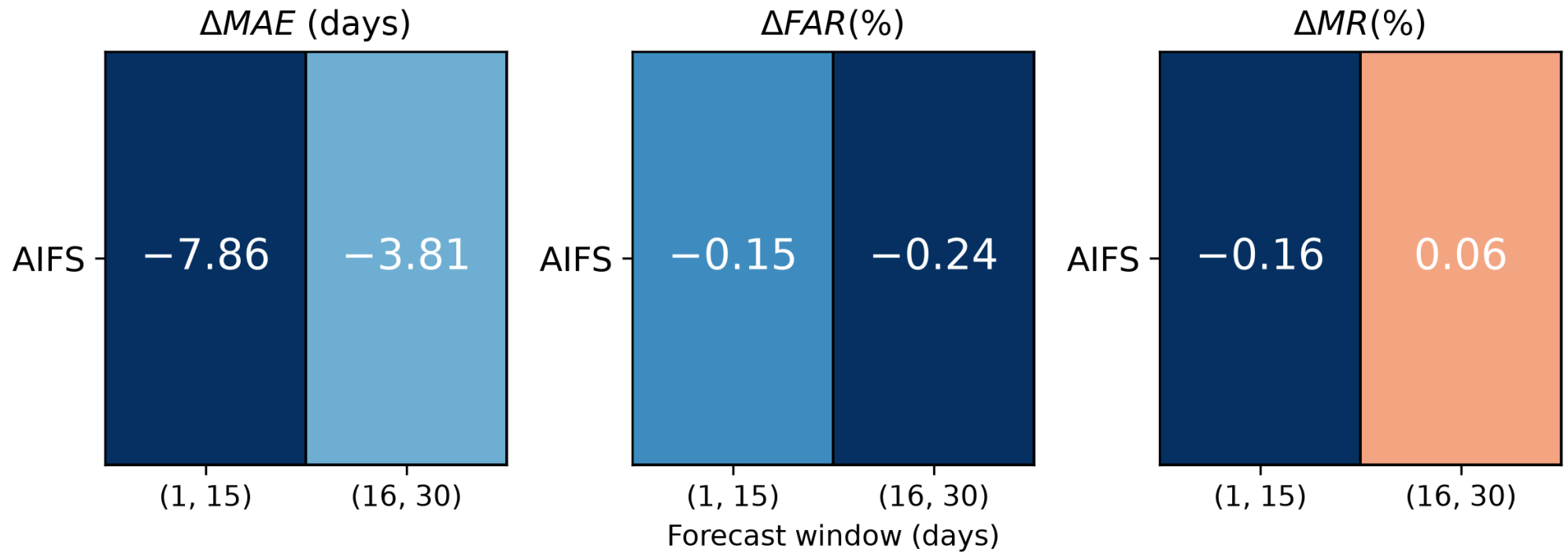
Detailed Metric Distributions

Model	Lead Window	Metric	Mean	Min	Median	P75	P90	Max
AIFS Single v2	Days 1-15	False Alarm Rate	27.042	0.000	14.706	45.114	75.000	100.000
AIFS Single v2	Days 1-15	Miss Rate	44.494	0.000	47.059	70.588	87.500	100.000
AIFS Single v2	Days 1-15	Mean Absolute Error	6.199	0.000	4.694	8.333	13.283	45.500
AIFS Single v2	Days 16-30	False Alarm Rate	38.984	0.000	25.000	75.000	100.000	100.000
AIFS Single v2	Days 16-30	Miss Rate	40.545	0.000	40.312	66.667	79.793	100.000
AIFS Single v2	Days 16-30	Mean Absolute Error	9.999	0.000	9.190	13.125	16.833	54.333
Traditional Climatology	Days 1-15	False Alarm Rate	42.178	0.000	37.500	60.000	80.000	100.000
Traditional Climatology	Days 1-15	Miss Rate	60.231	0.000	62.500	83.333	100.000	100.000
Traditional Climatology	Days 1-15	Mean Absolute Error	14.059	0.000	10.250	16.000	25.900	97.000
Traditional Climatology	Days 16-30	False Alarm Rate	63.025	0.000	72.727	100.000	100.000	100.000
Traditional Climatology	Days 16-30	Miss Rate	34.211	0.000	31.034	48.276	65.152	100.000
Traditional Climatology	Days 16-30	Mean Absolute Error	13.809	0.000	11.333	17.688	25.450	85.000
FuXi	Days 1-15	False Alarm Rate	16.619	0.000	0.000	23.190	61.750	100.000
FuXi	Days 1-15	Miss Rate	69.188	0.000	87.500	100.000	100.000	100.000
FuXi	Days 1-15	Mean Absolute Error	5.123	0.000	3.500	7.031	10.393	46.000
FuXi	Days 16-30	False Alarm Rate	16.266	0.000	0.000	25.000	60.000	100.000
FuXi	Days 16-30	Miss Rate	68.135	0.000	86.667	100.000	100.000	100.000
FuXi	Days 16-30	Mean Absolute Error	10.497	0.000	9.000	14.833	19.700	42.000
GraphCast	Days 1-15	False Alarm Rate	28.914	0.000	18.750	50.000	80.000	100.000
GraphCast	Days 1-15	Miss Rate	43.893	0.000	37.500	75.000	100.000	100.000
GraphCast	Days 1-15	Mean Absolute Error	6.160	0.000	4.458	8.375	12.562	59.750
GraphCast	Days 16-30	False Alarm Rate	19.146	0.000	0.000	25.000	75.000	100.000
GraphCast	Days 16-30	Miss Rate	44.970	0.000	40.000	78.571	100.000	100.000
GraphCast	Days 16-30	Mean Absolute Error	7.631	0.000	6.333	10.000	15.000	40.000

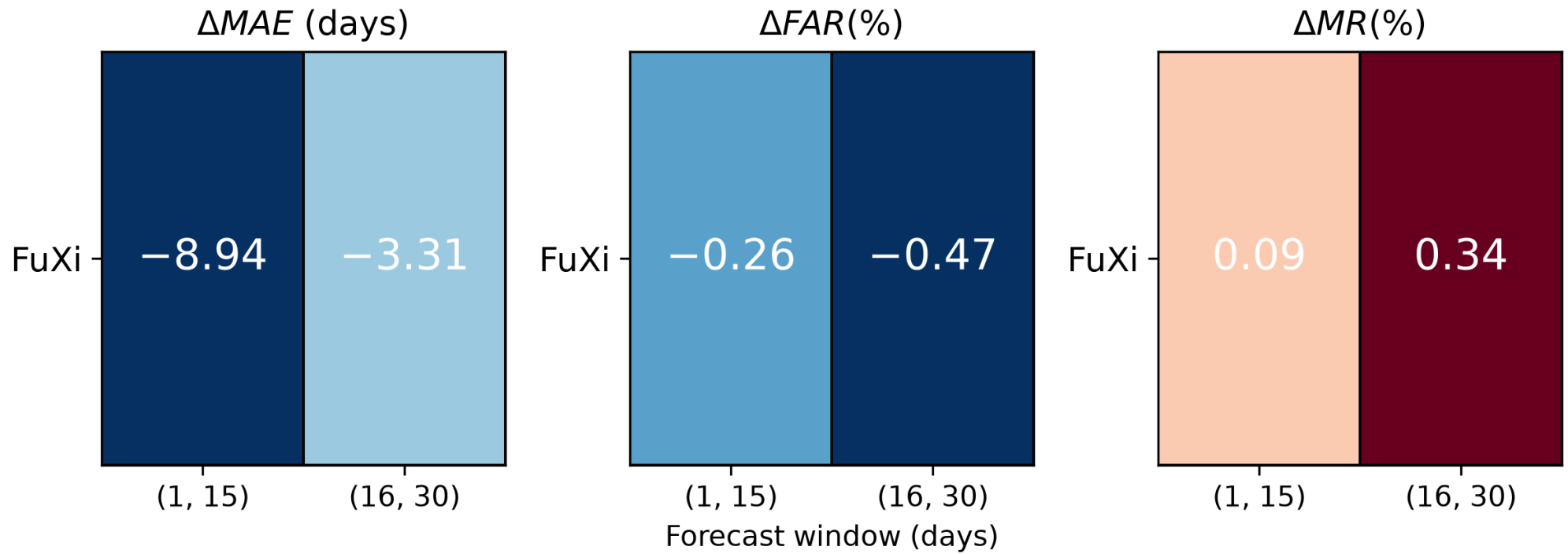
Benchmark Output Figures

Deterministic AI

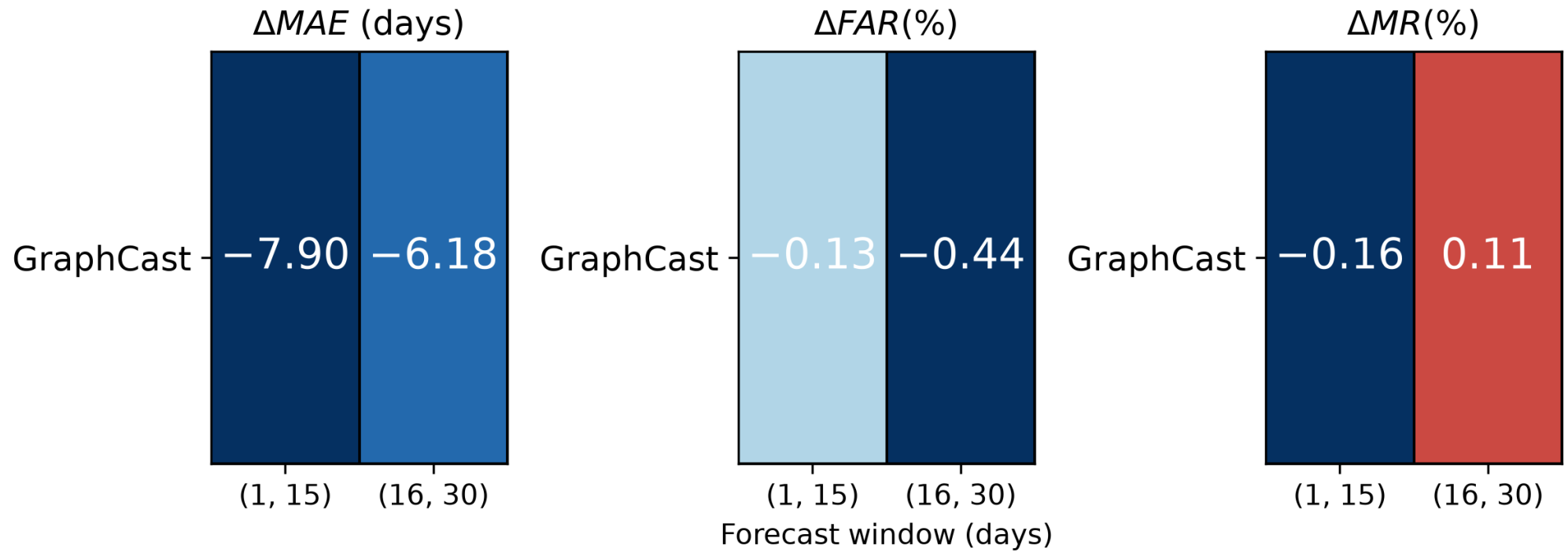
Portrait Panel (4)



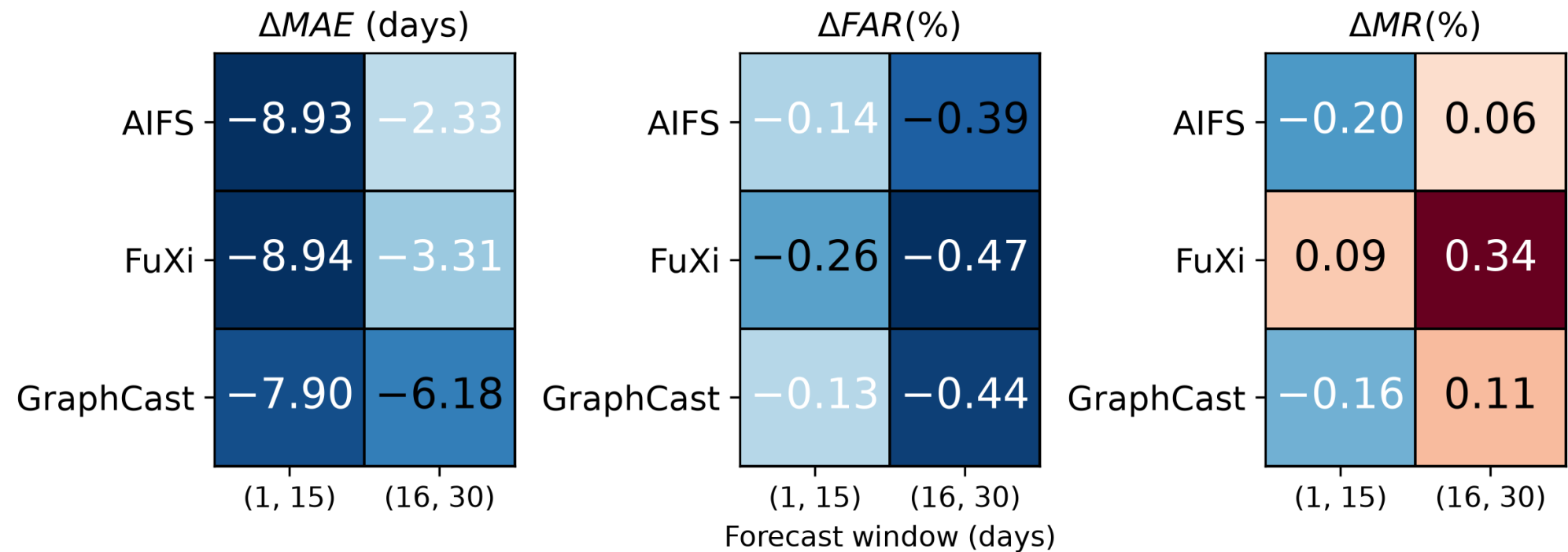
Panel Portrait: MAE / FAR / MR - AIFS (30-day)



Panel Portrait MAE / FAR / MR - FuXi - 30-day

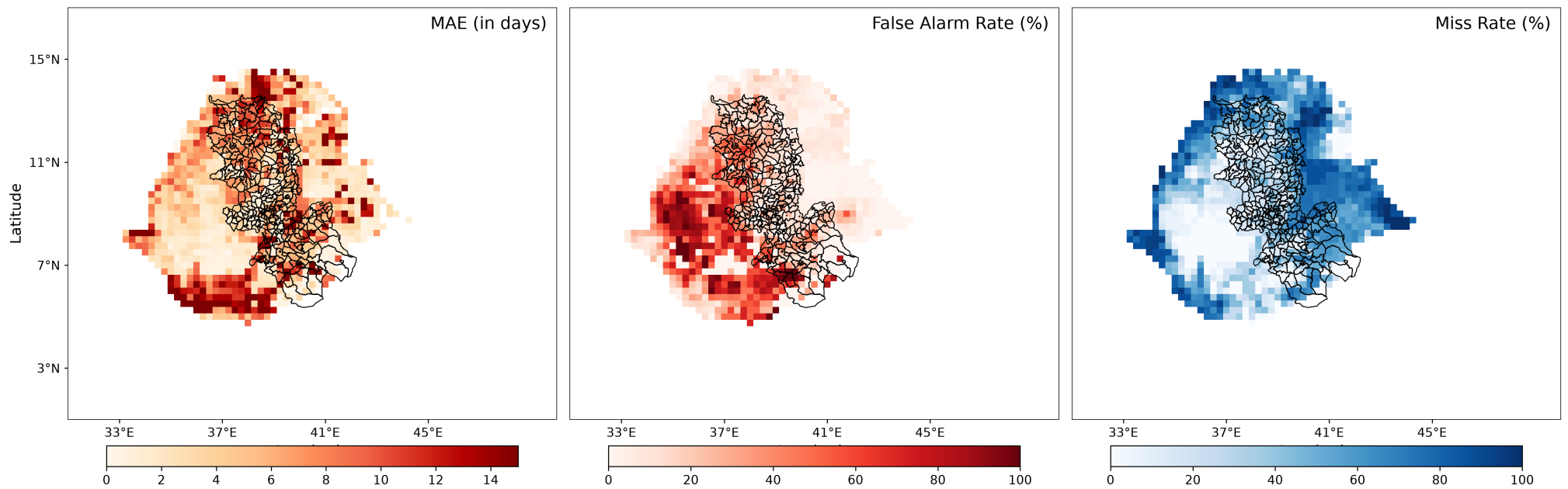


Panel Portrait MAE / FAR / MR - GraphCast - 30-day

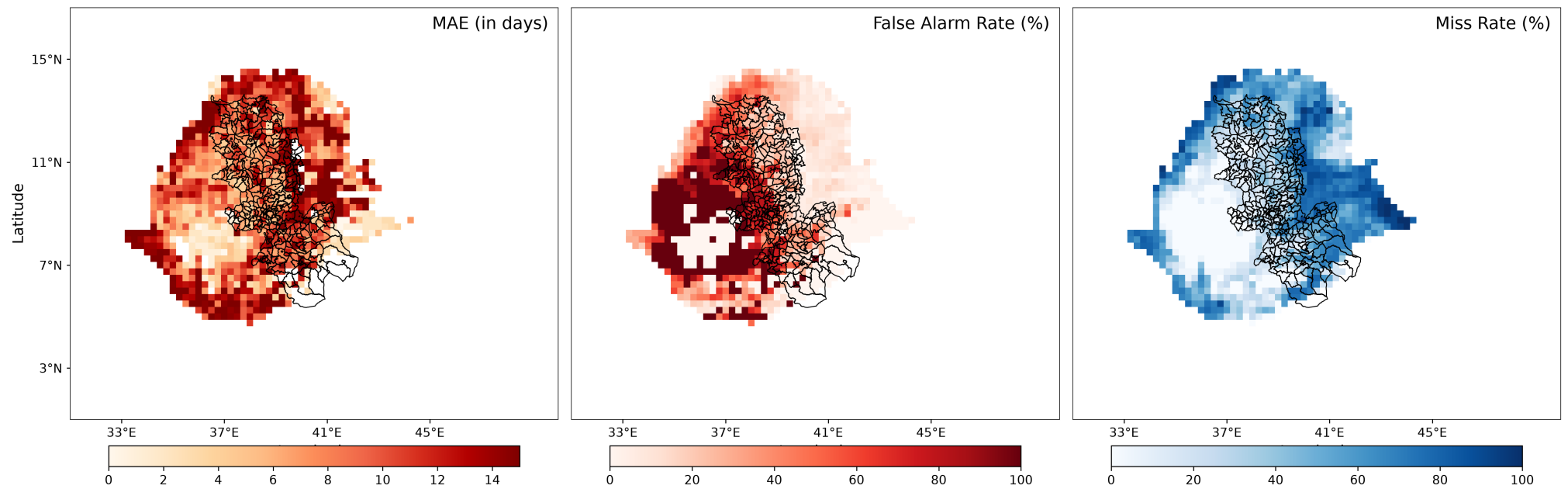


Panel Portrait: MAE / FAR / MR - AIFS (30-day)

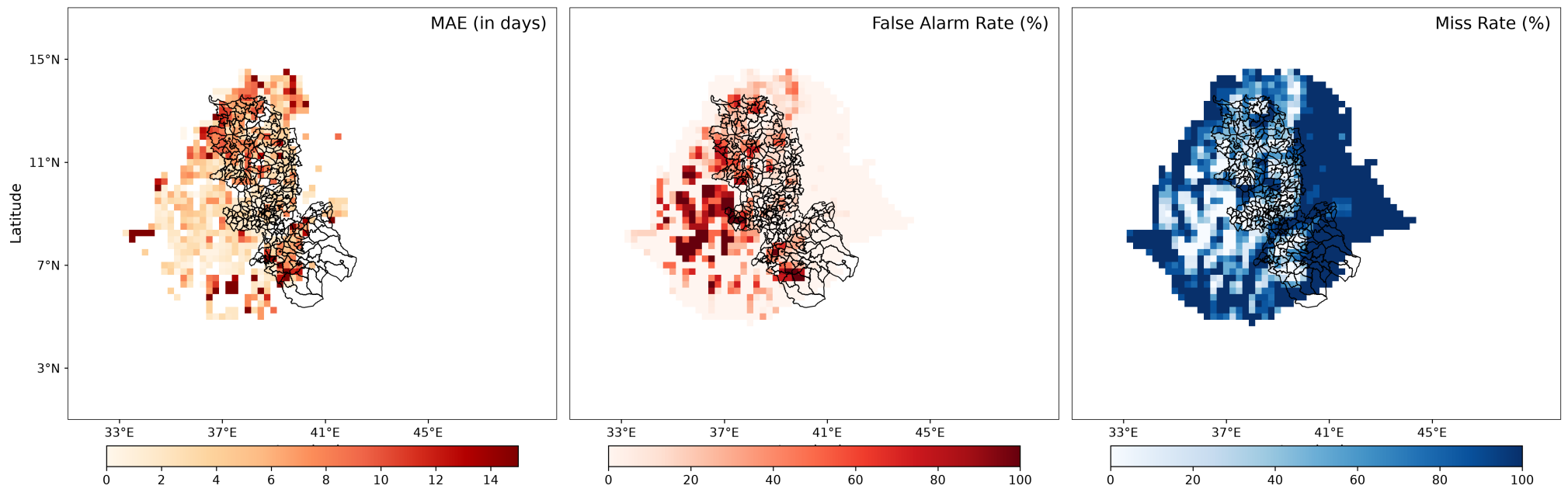
Spatial Metrics (MAE / FAR / Miss Rate) (12)



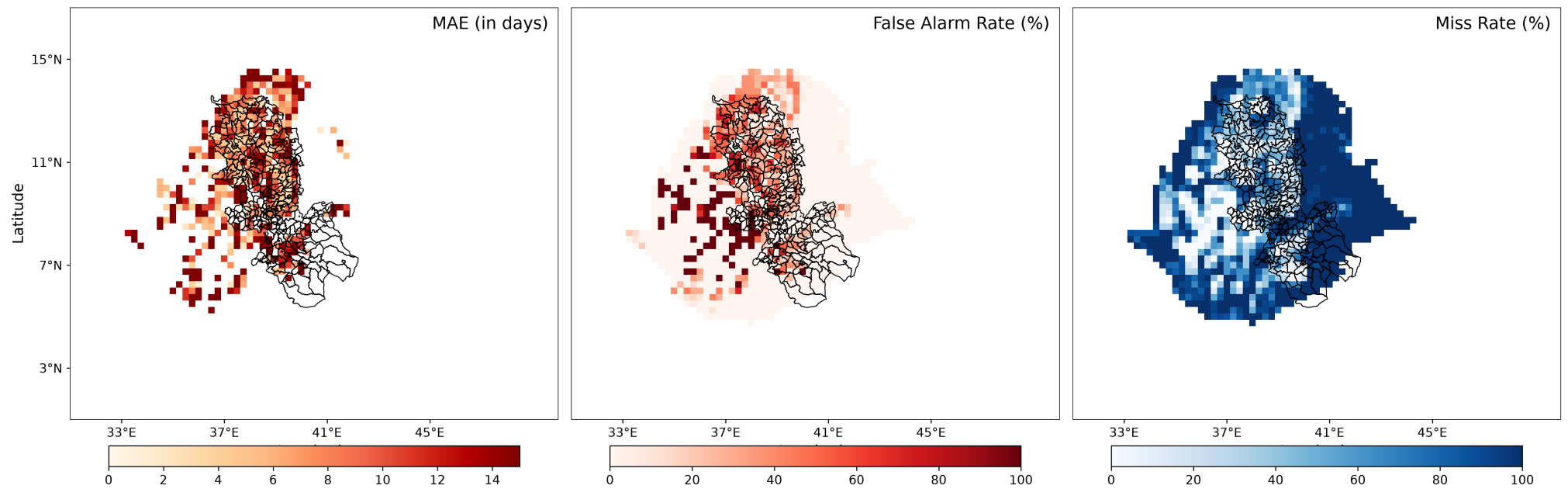
Spatial Metrics - AIFS - Days 1-15



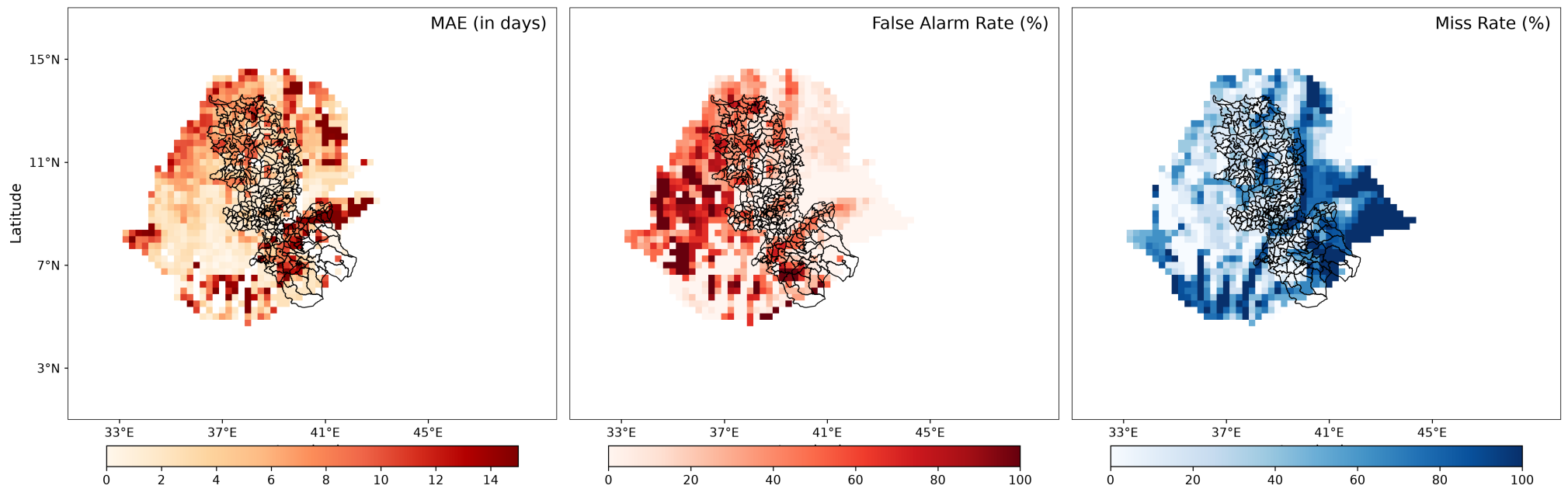
Spatial Metrics - AIFS - Days 16-30



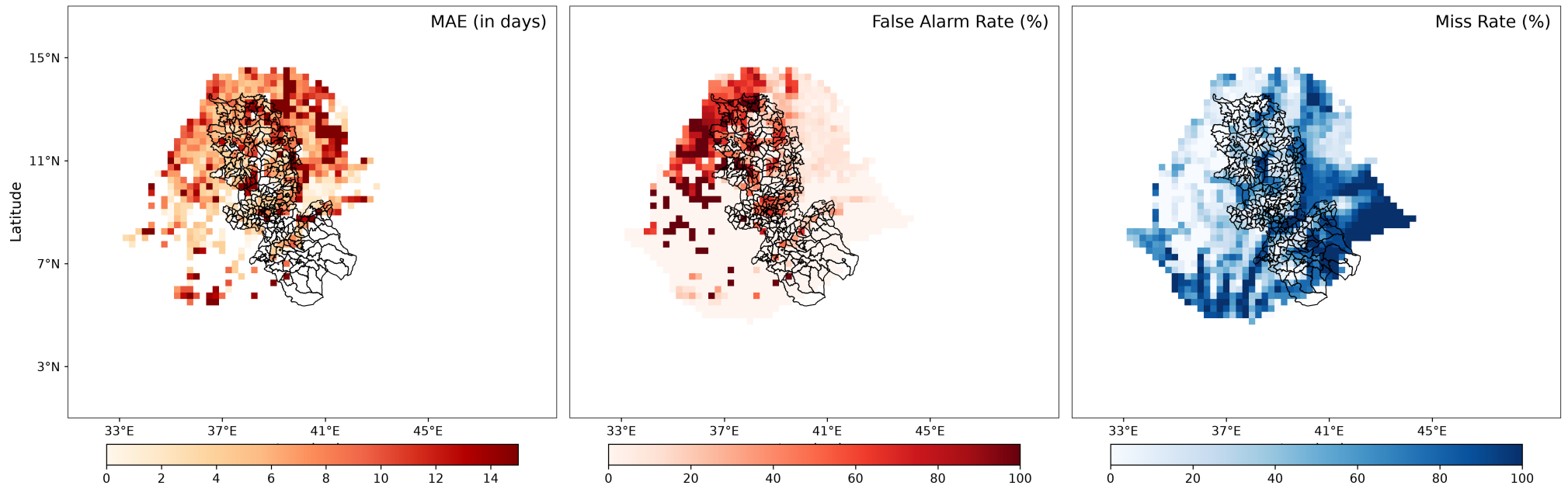
Spatial Metrics - FuXi - Days 1-15



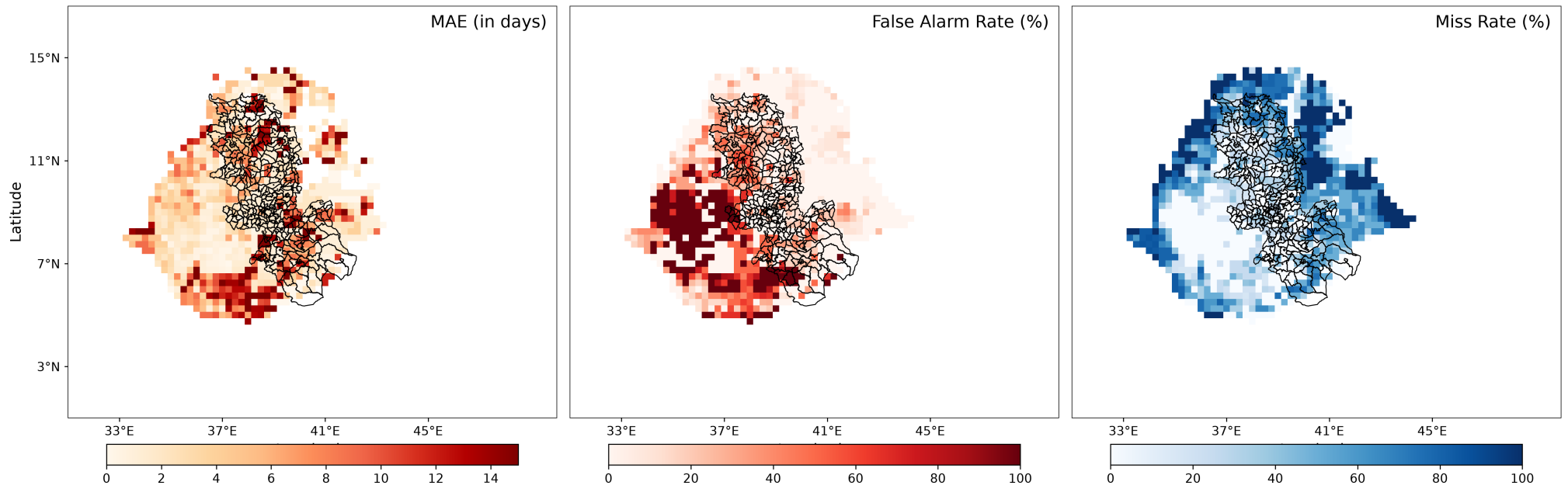
Spatial Metrics - FuXi - Days 16-30



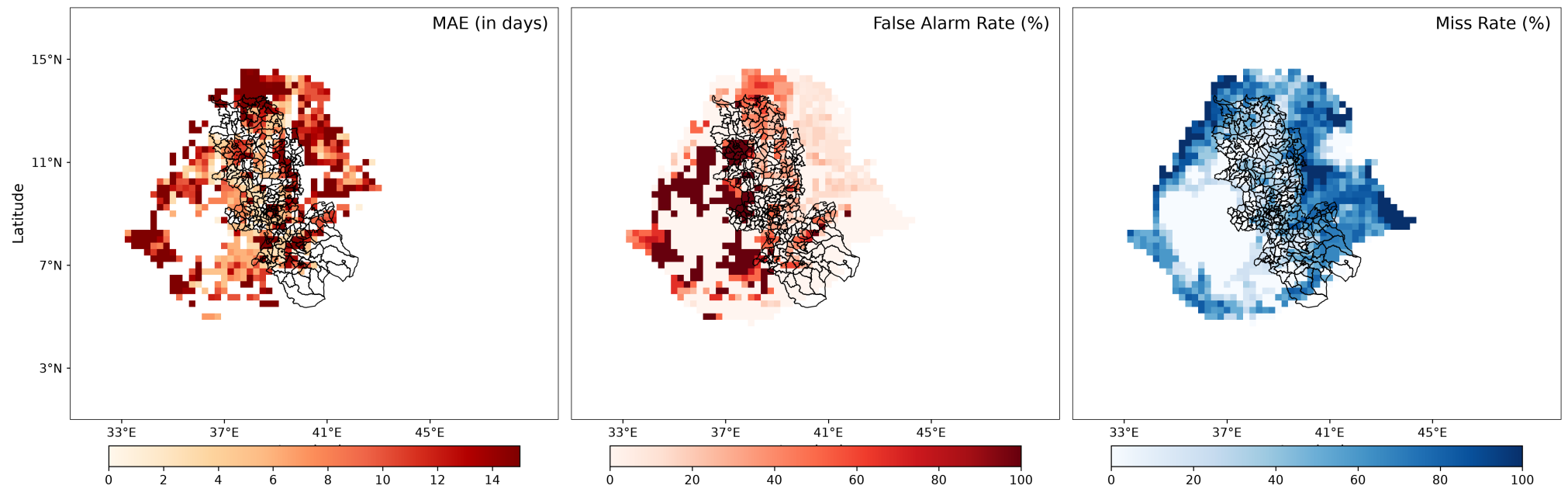
Spatial Metrics - GraphCast - Days 1-15



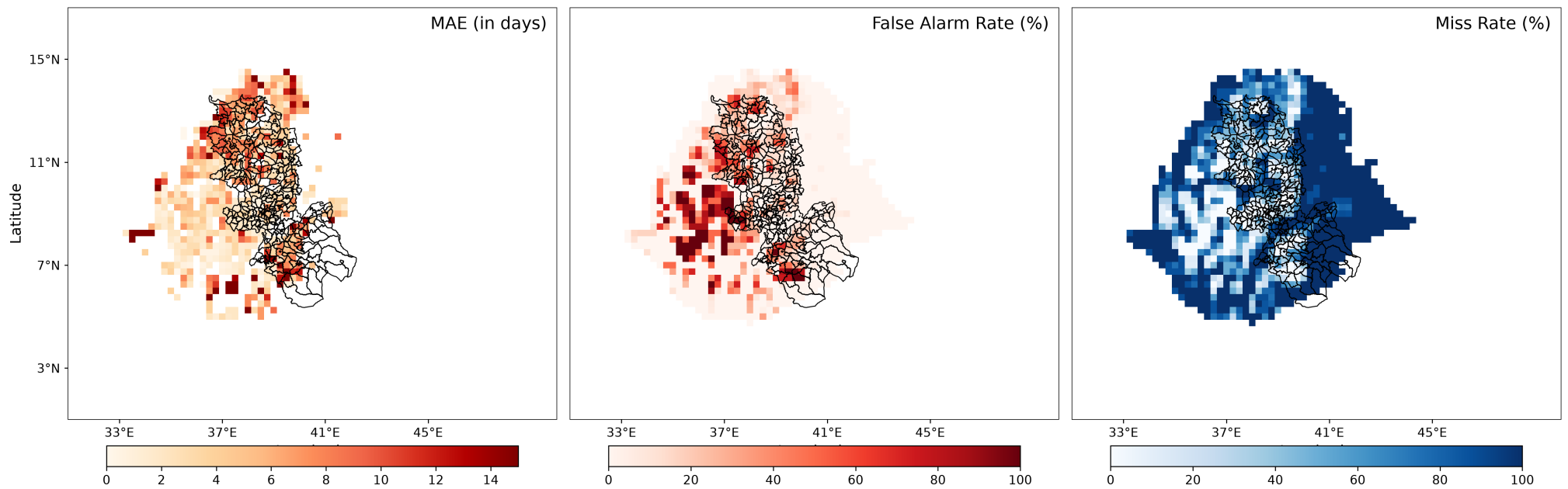
Spatial Metrics - GraphCast - Days 16-30



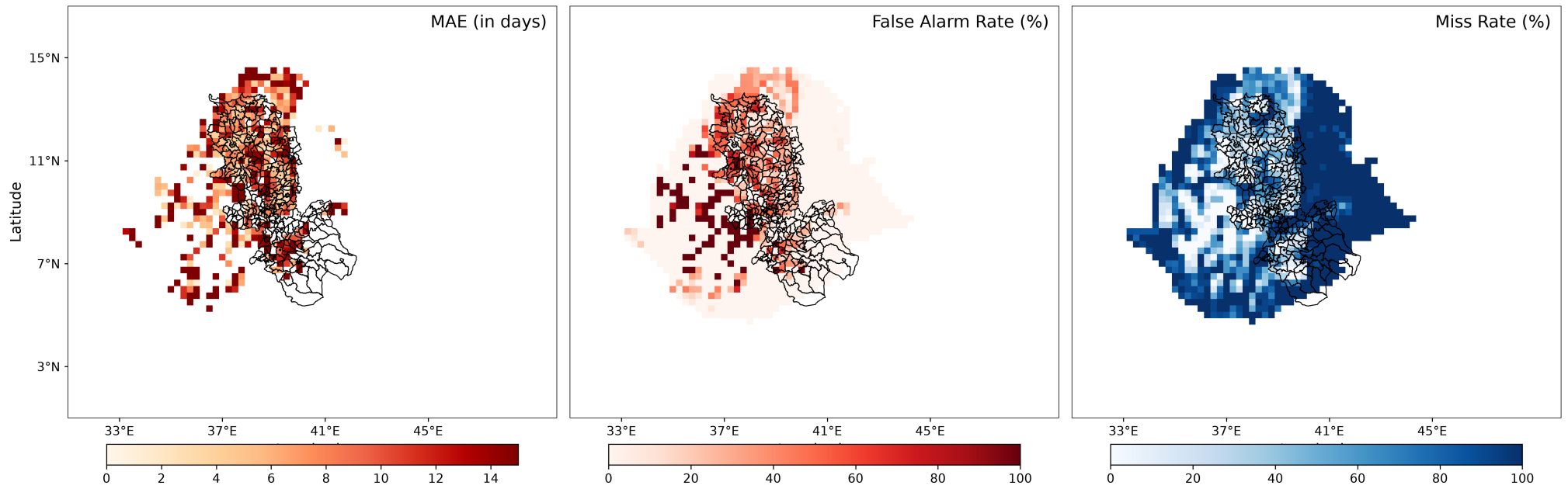
Spatial Metrics - AIFS - Days 1-15



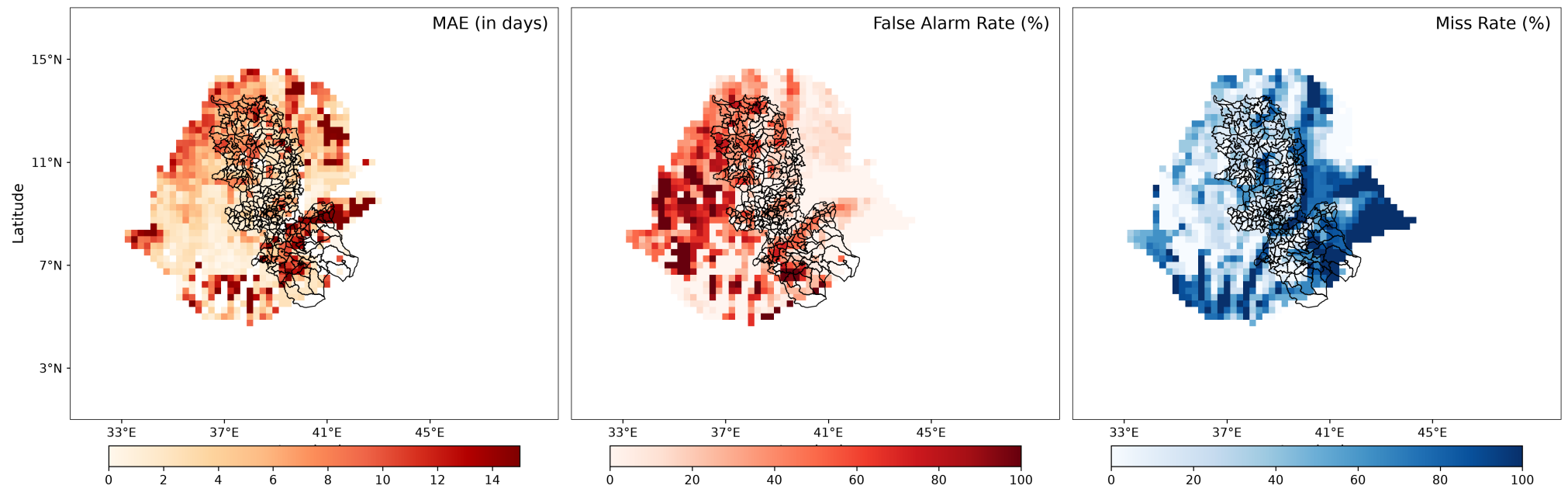
Spatial Metrics - AIFS - Days 16-30



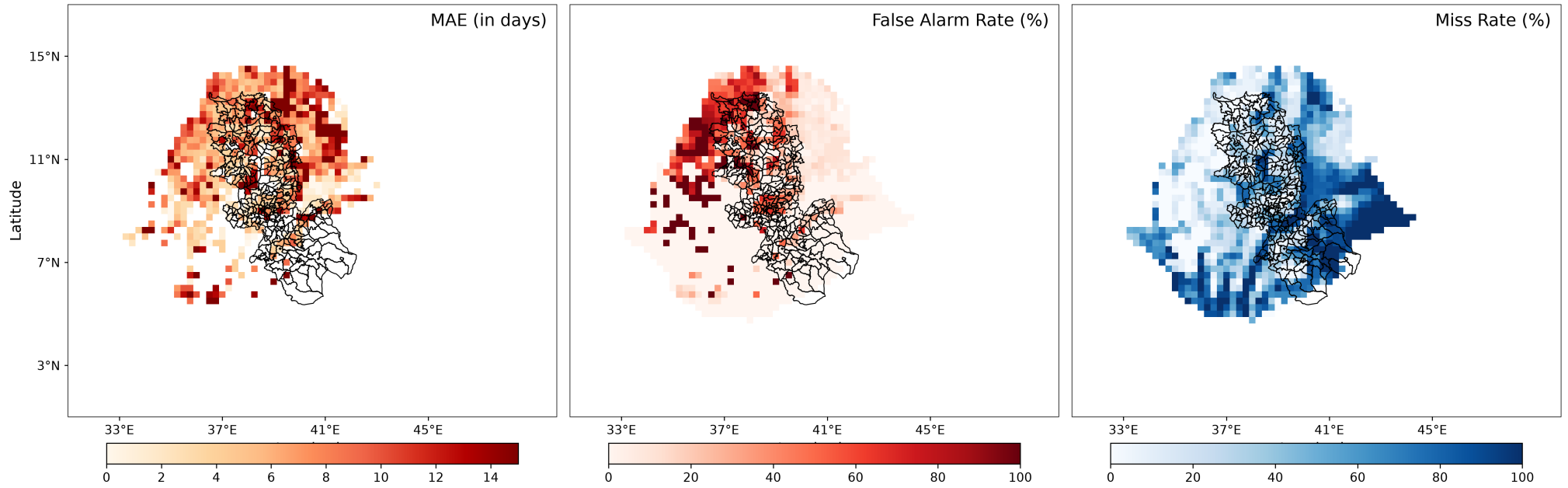
Spatial Metrics - FuXi - Days 1-15



Spatial Metrics - FuXi - Days 16-30



Spatial Metrics - GraphCast - Days 1-15

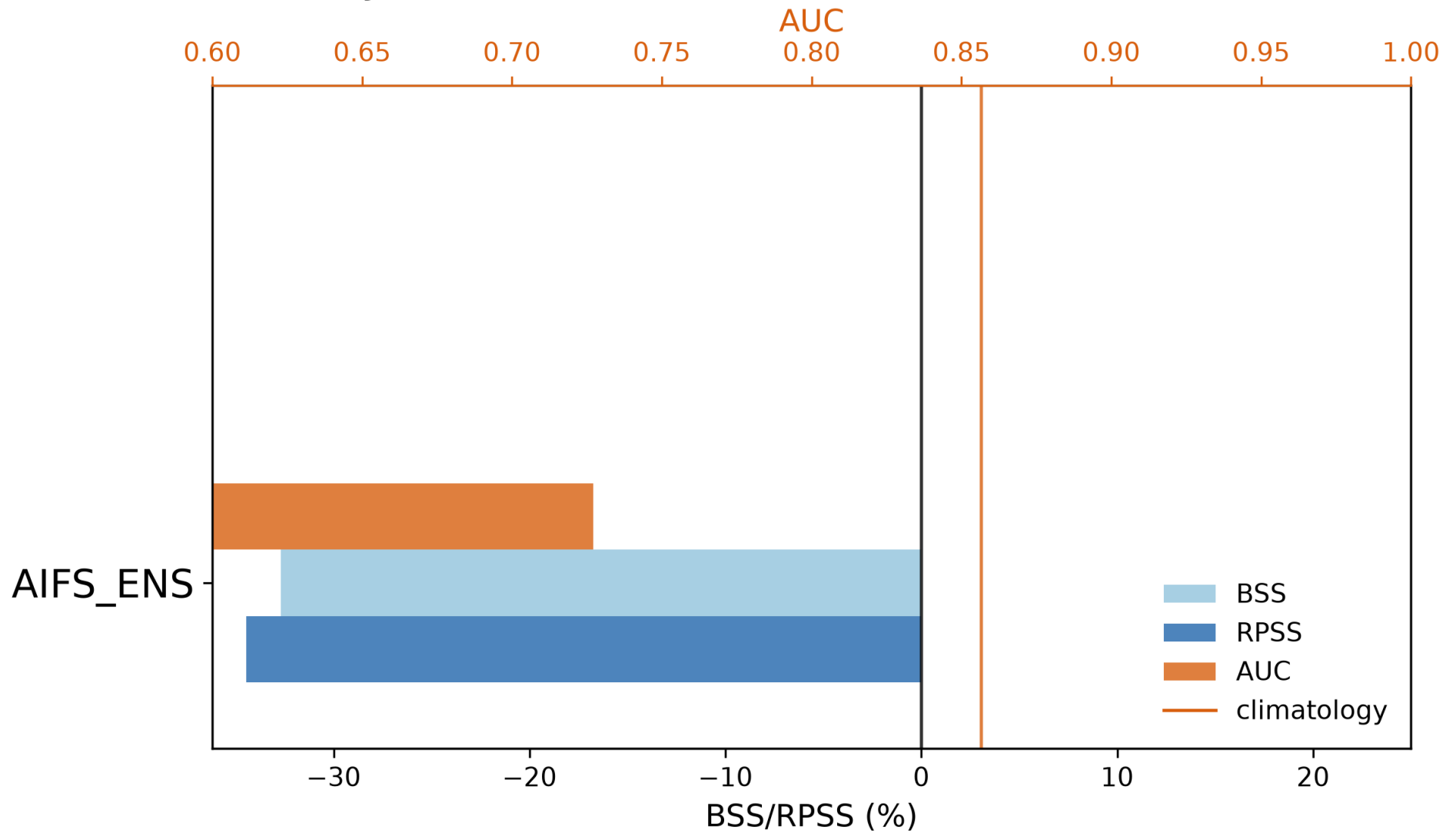


Spatial Metrics - GraphCast - Days 16-30

Probabilistic AI

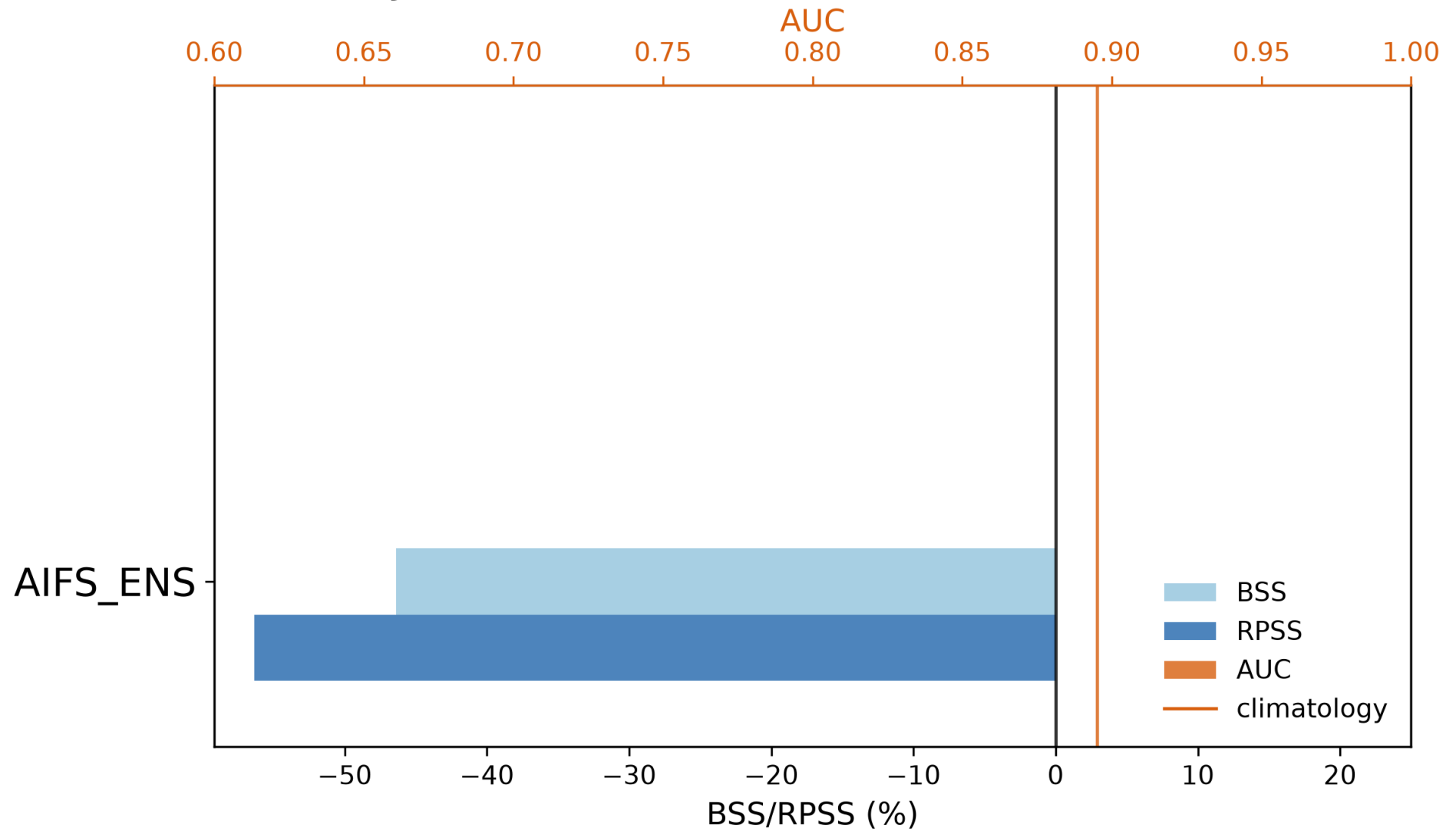
Other (6)

1-15 day forecast



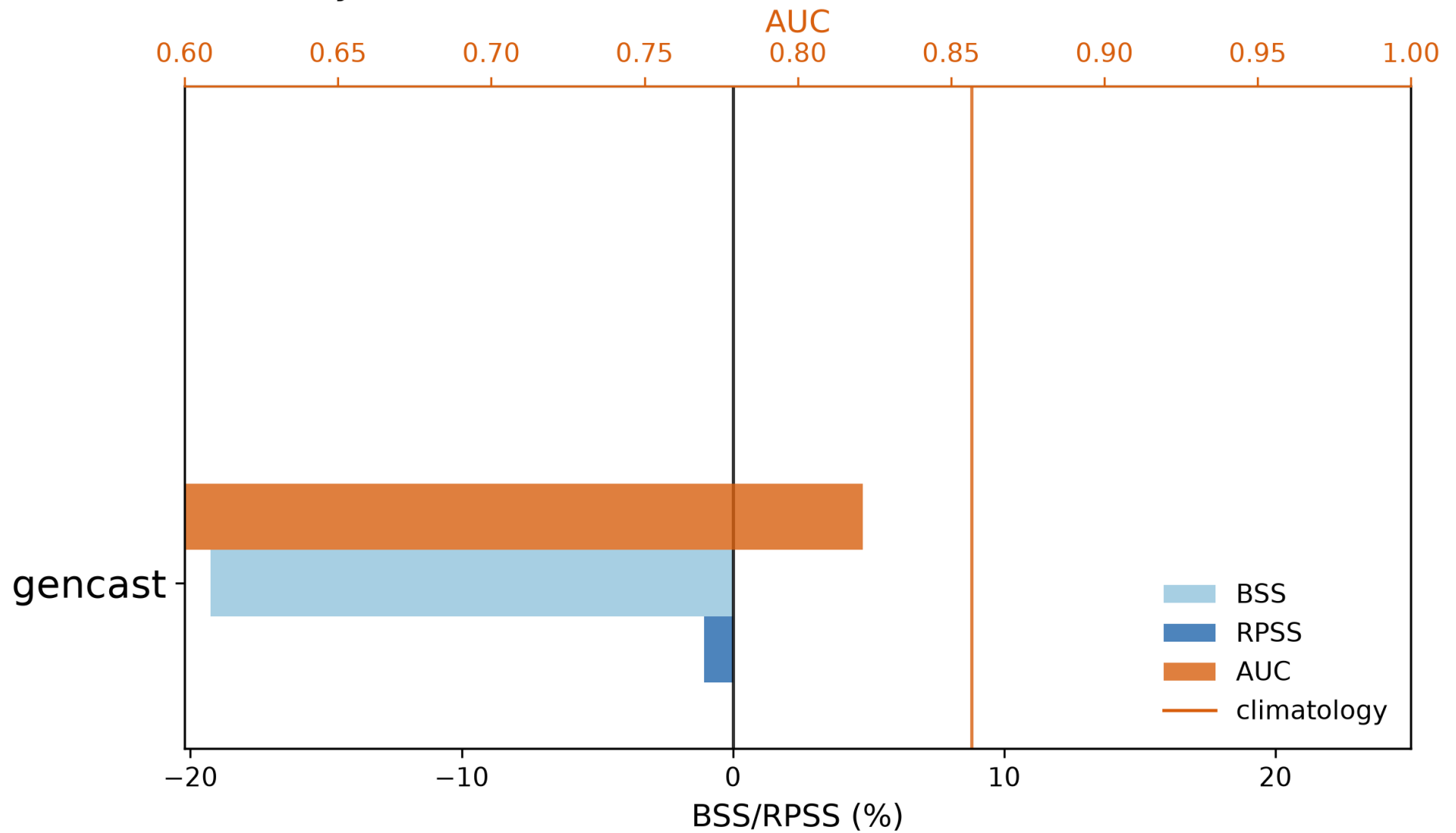
Panel Bar Bss Rpss Auc 1-15 - Days 1-15

16-30 day forecast



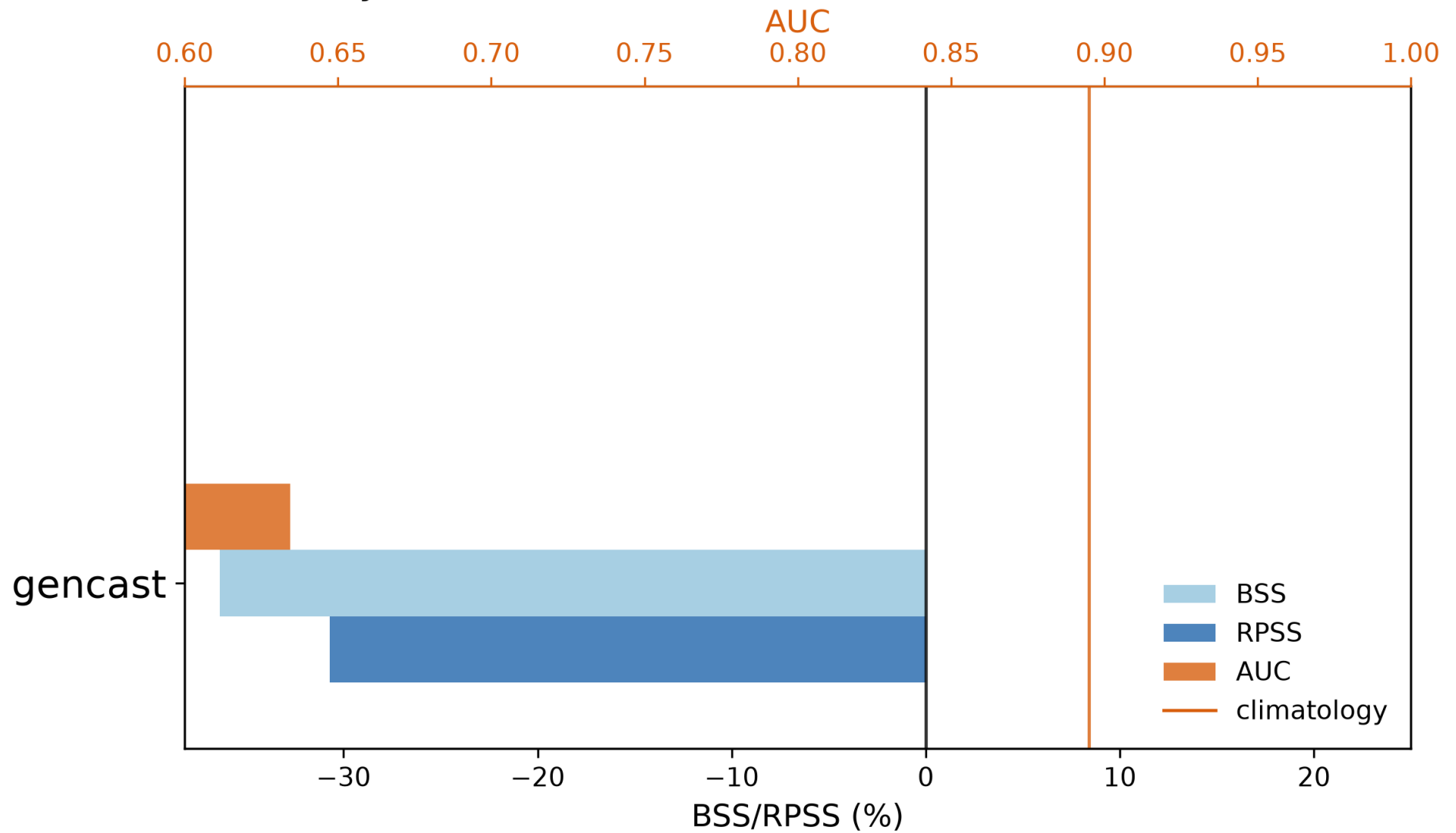
Panel Bar Bss Rpss Auc 16-30 - Days 16-30

1-15 day forecast



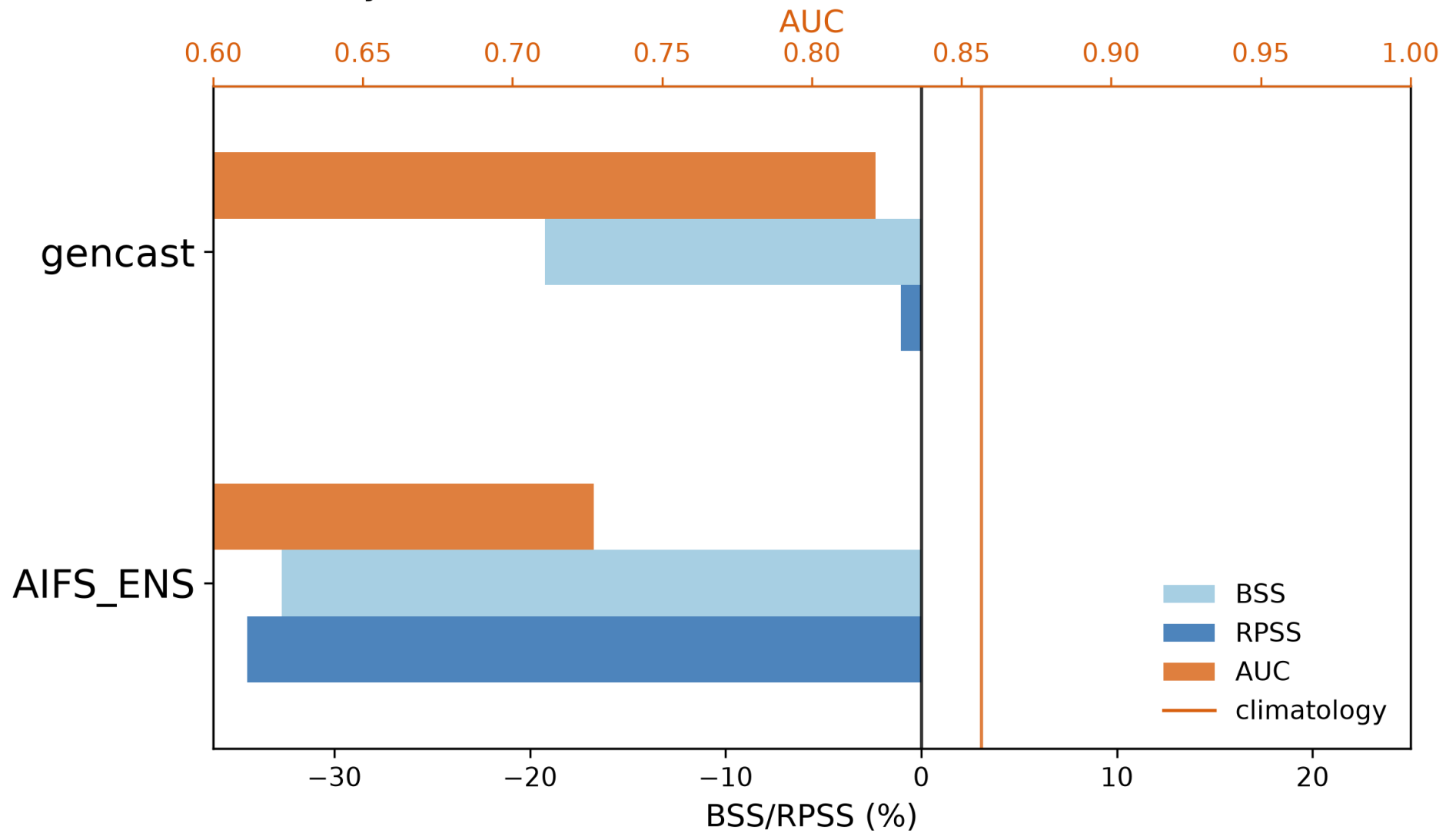
Panel Bar Bss Rpss Auc 1-15 - Days 1-15

16-30 day forecast



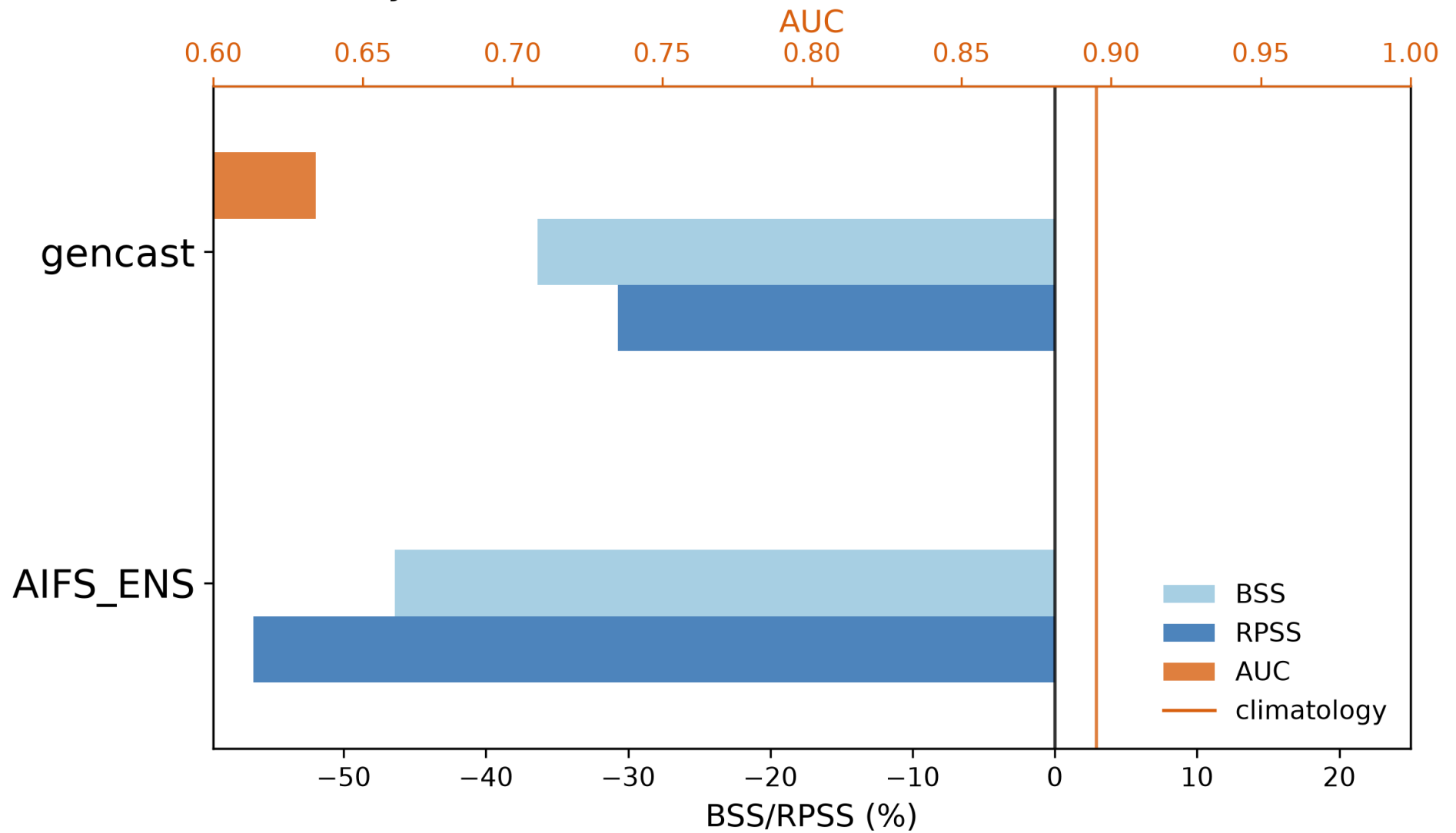
Panel Bar Bss Rpss Auc 16-30 - Days 16-30

1-15 day forecast

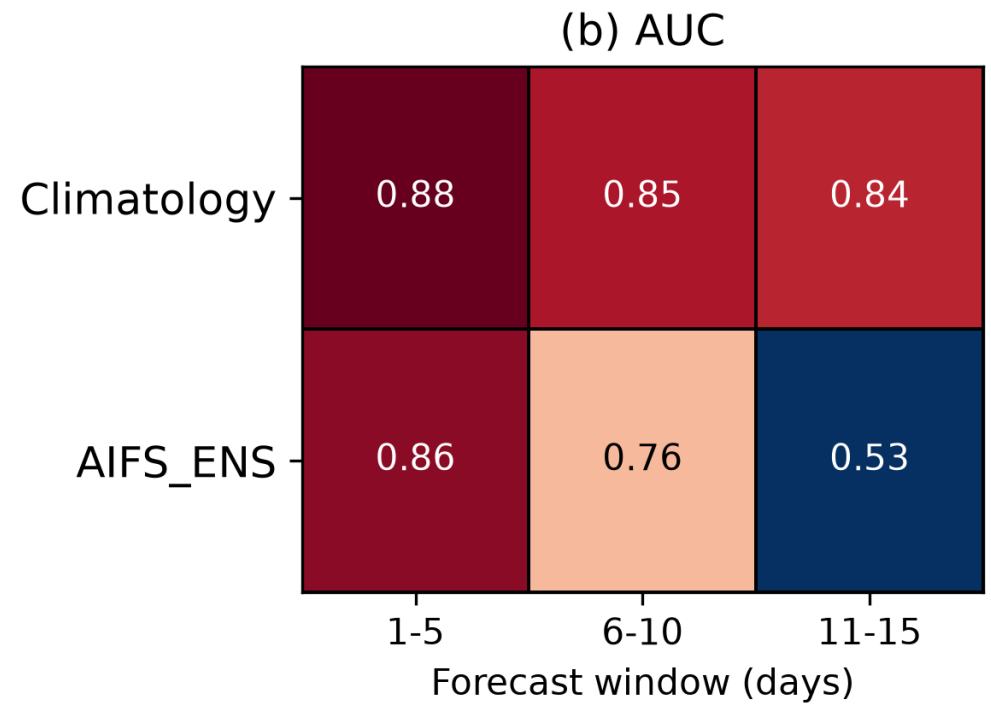
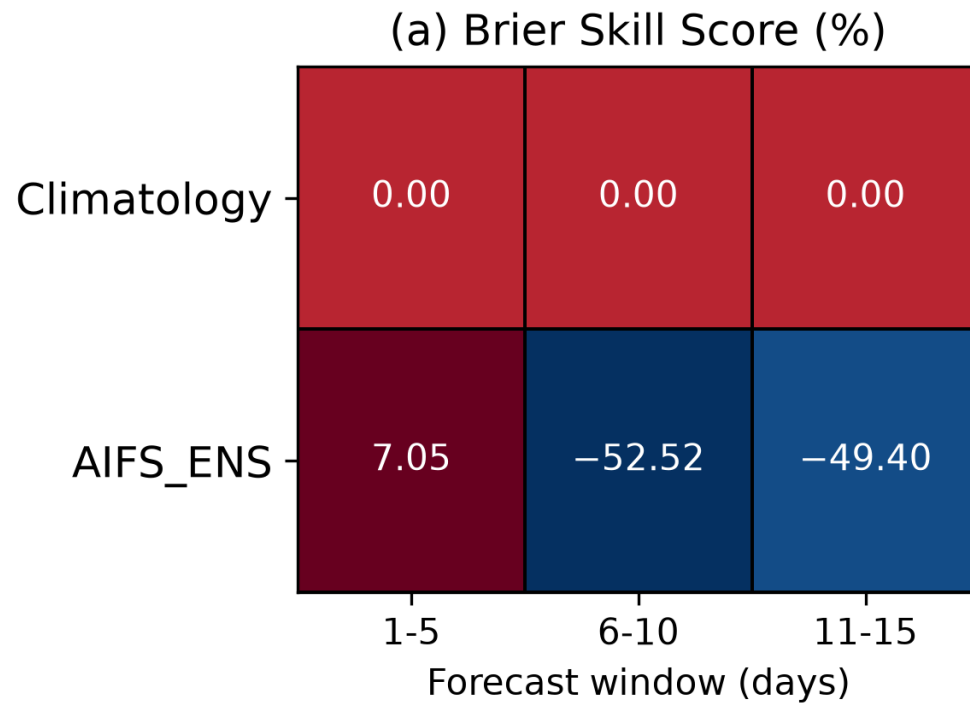


Panel Bar Bss Rpss Auc 1-15 - Days 1-15

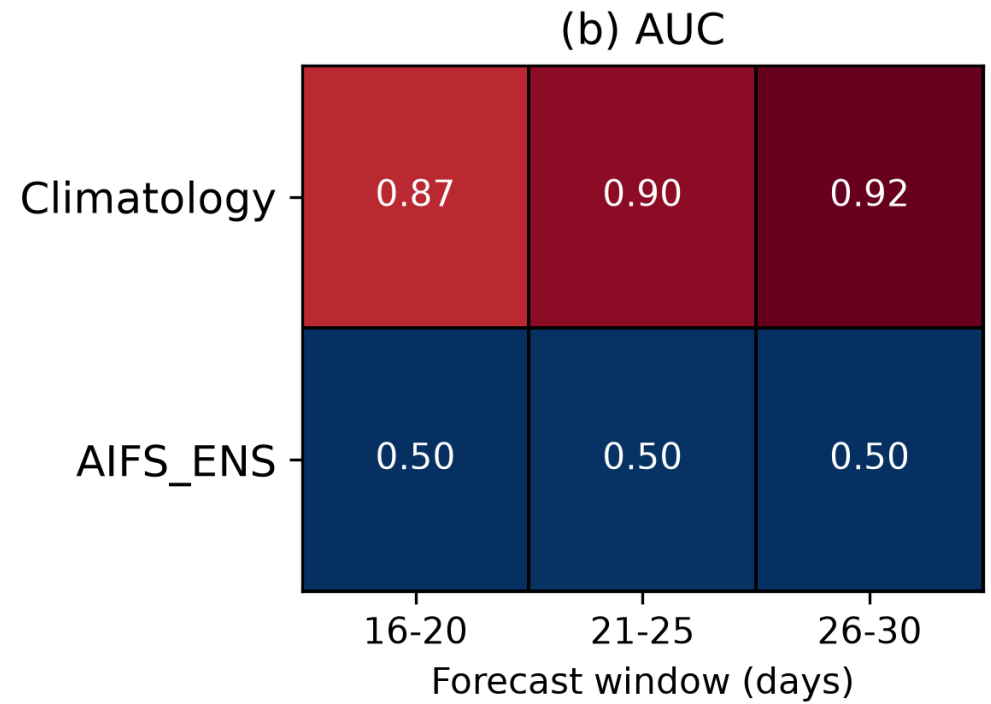
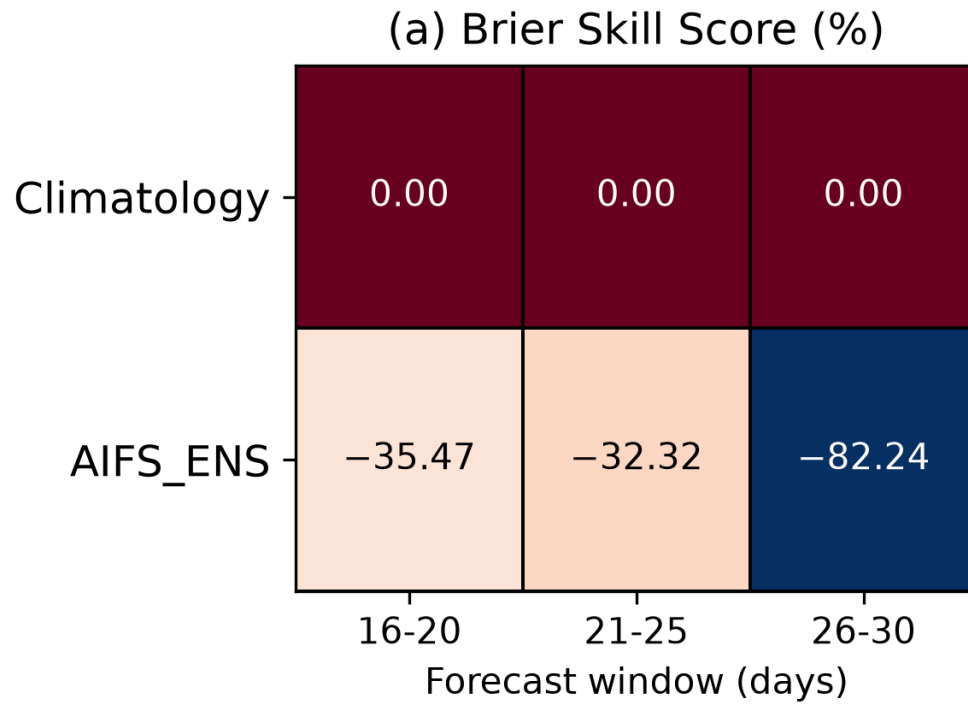
16-30 day forecast



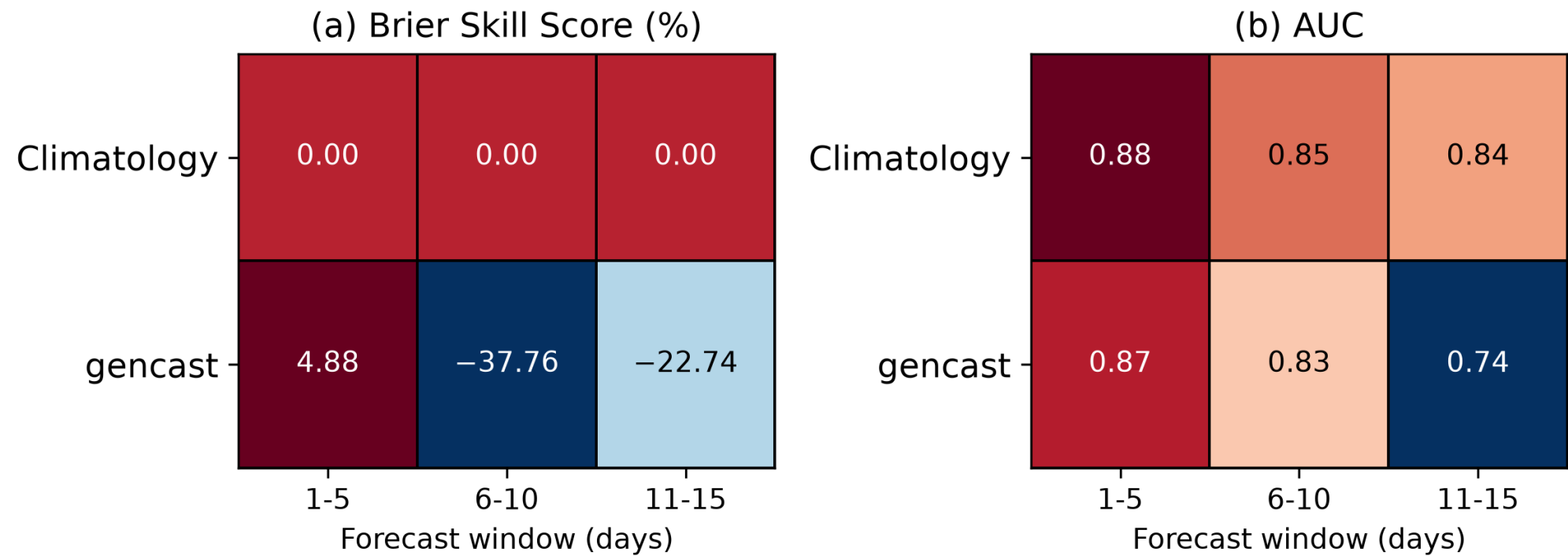
Panel Bar Bss Rpss Auc 16-30 - Days 16-30



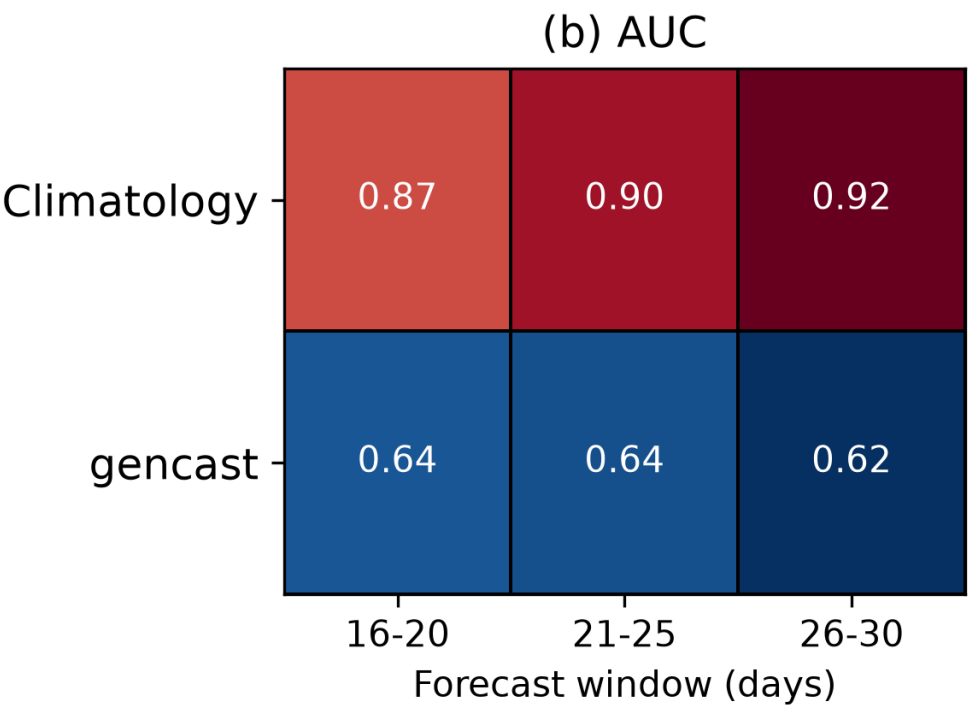
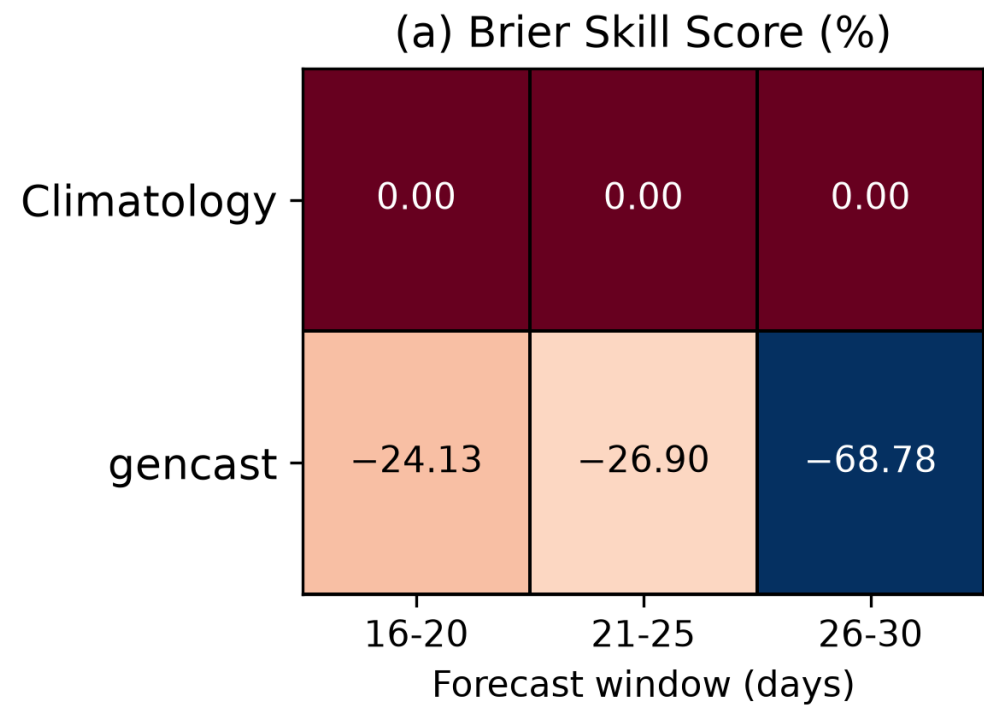
Panel Portrait - Days 1-15



Panel Portrait - Days 16-30

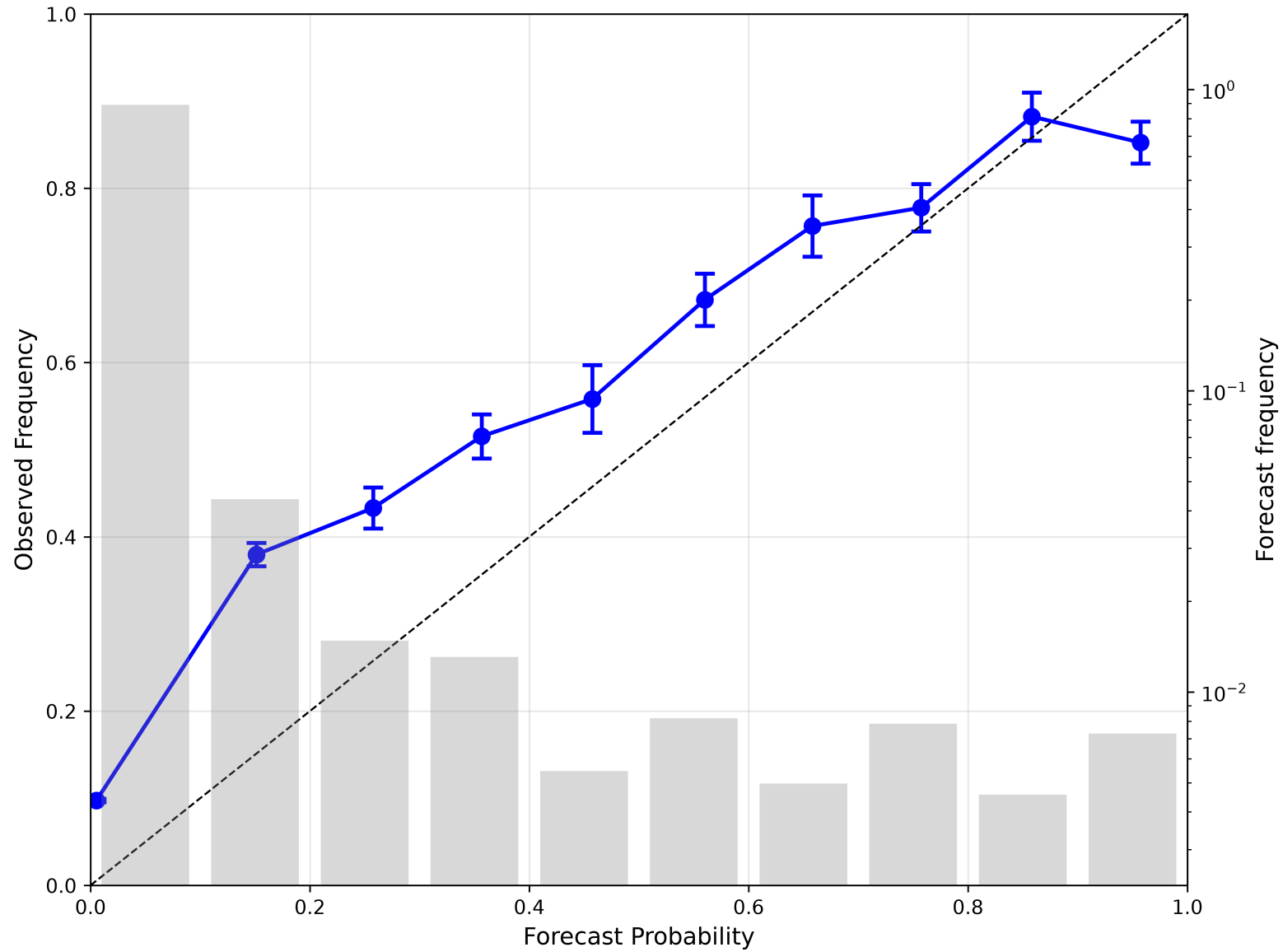


Panel Portrait - Days 1-15

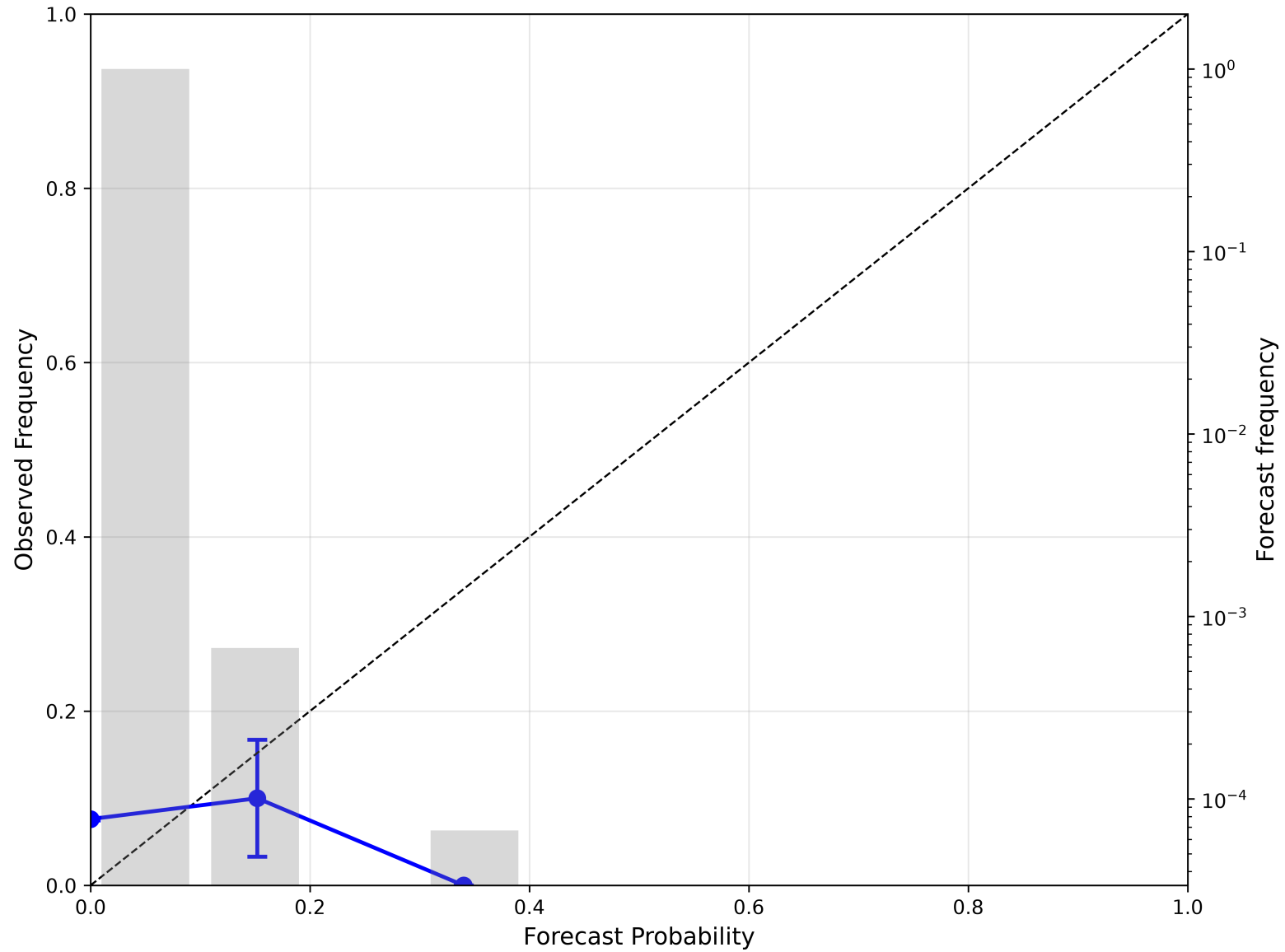


Panel Portrait - Days 16-30

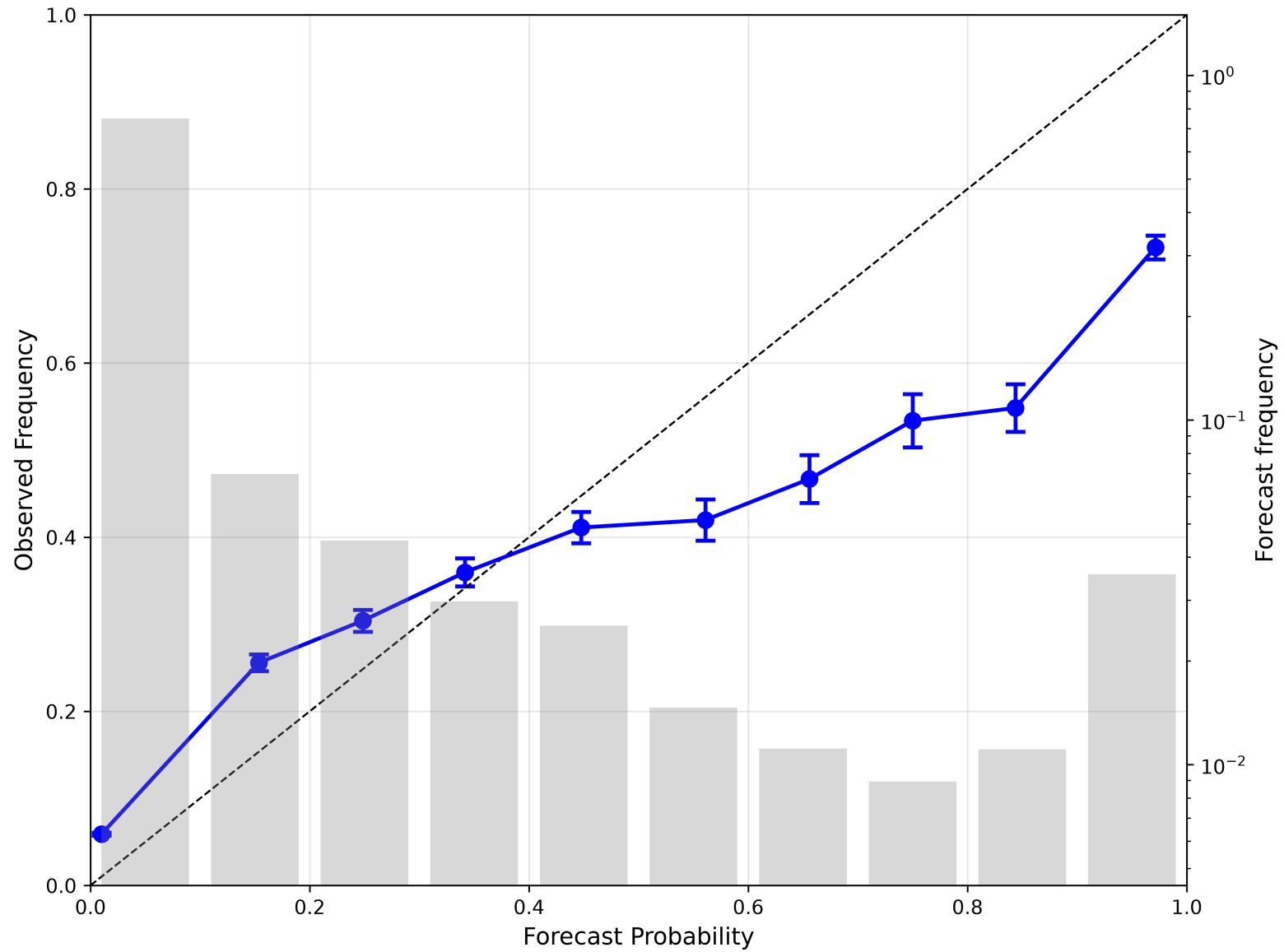
Reliability Diagram (8)



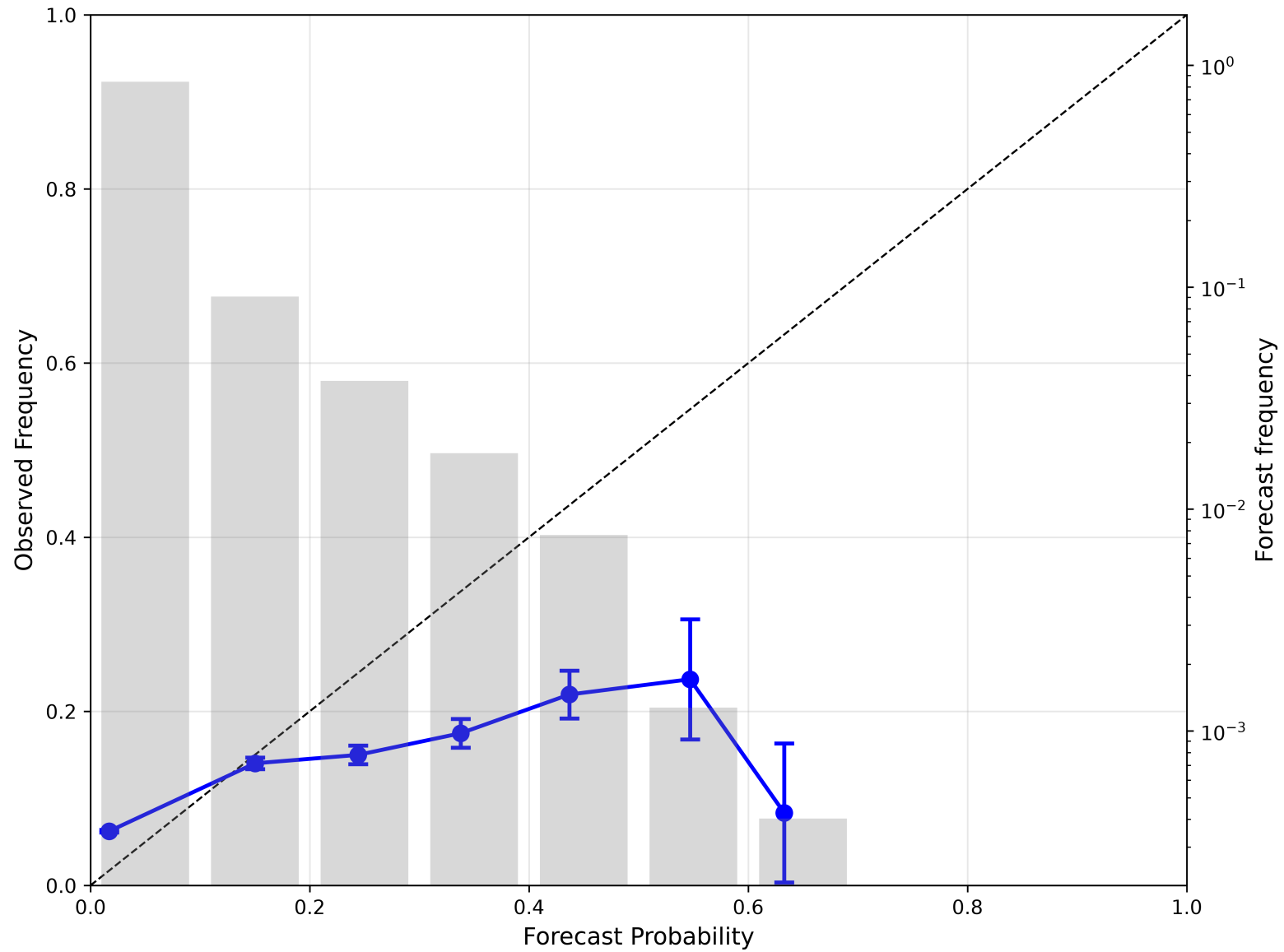
Reliability Diagram - AIFS Ensemble - Days 1-15



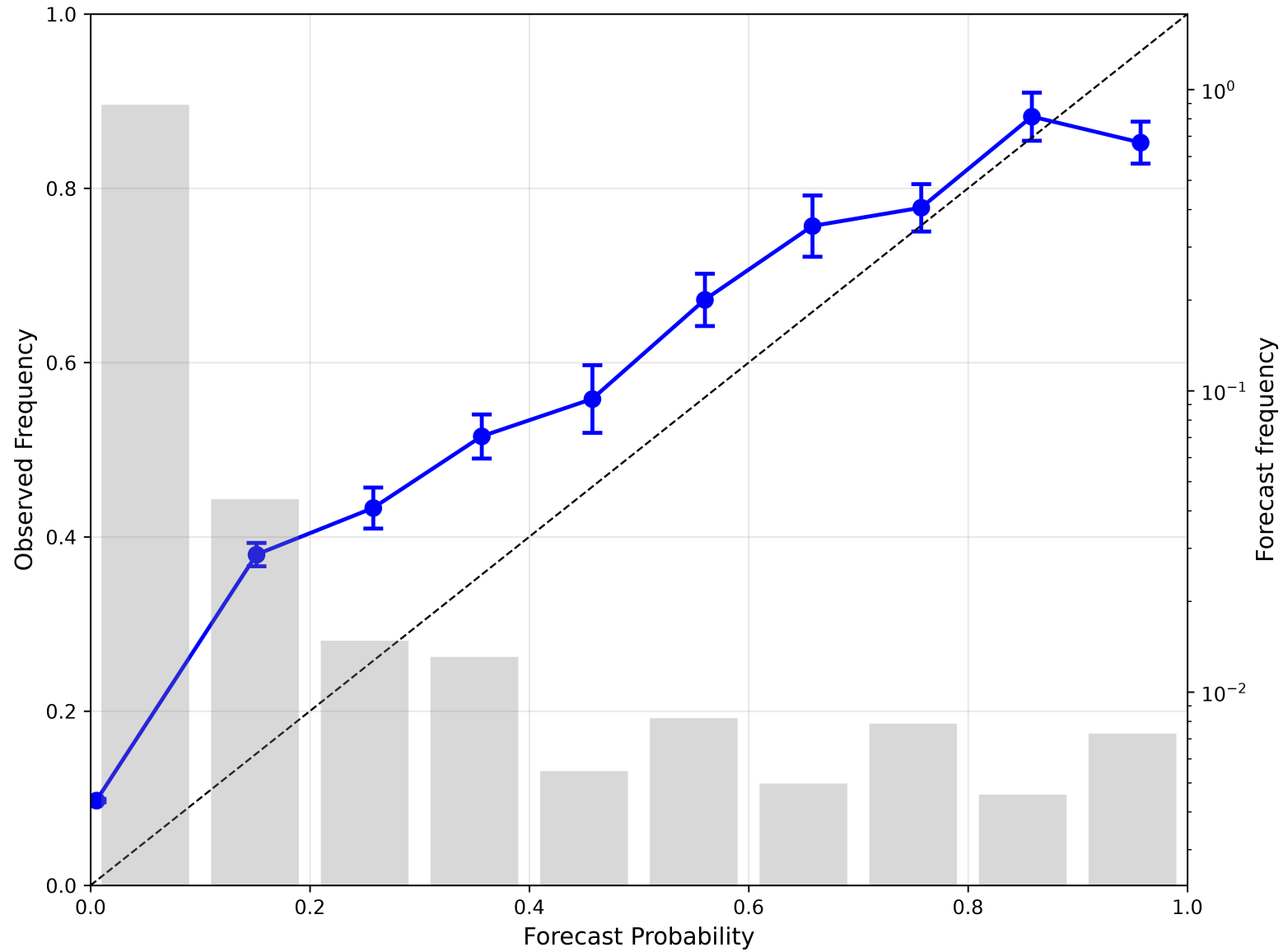
Reliability Diagram - AIFS Ensemble - Days 16-30



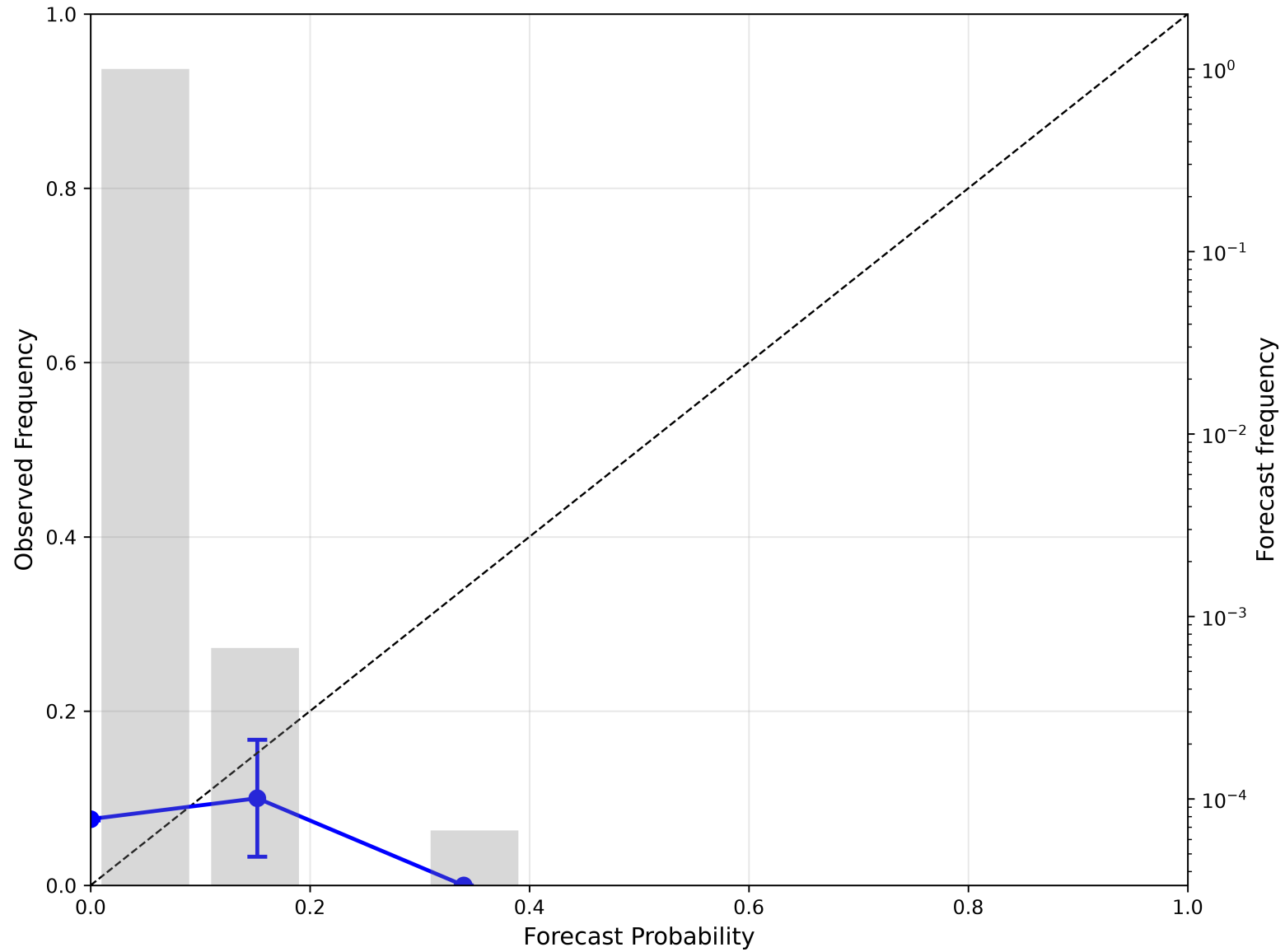
Reliability Diagram - GenCast - Days 1-15



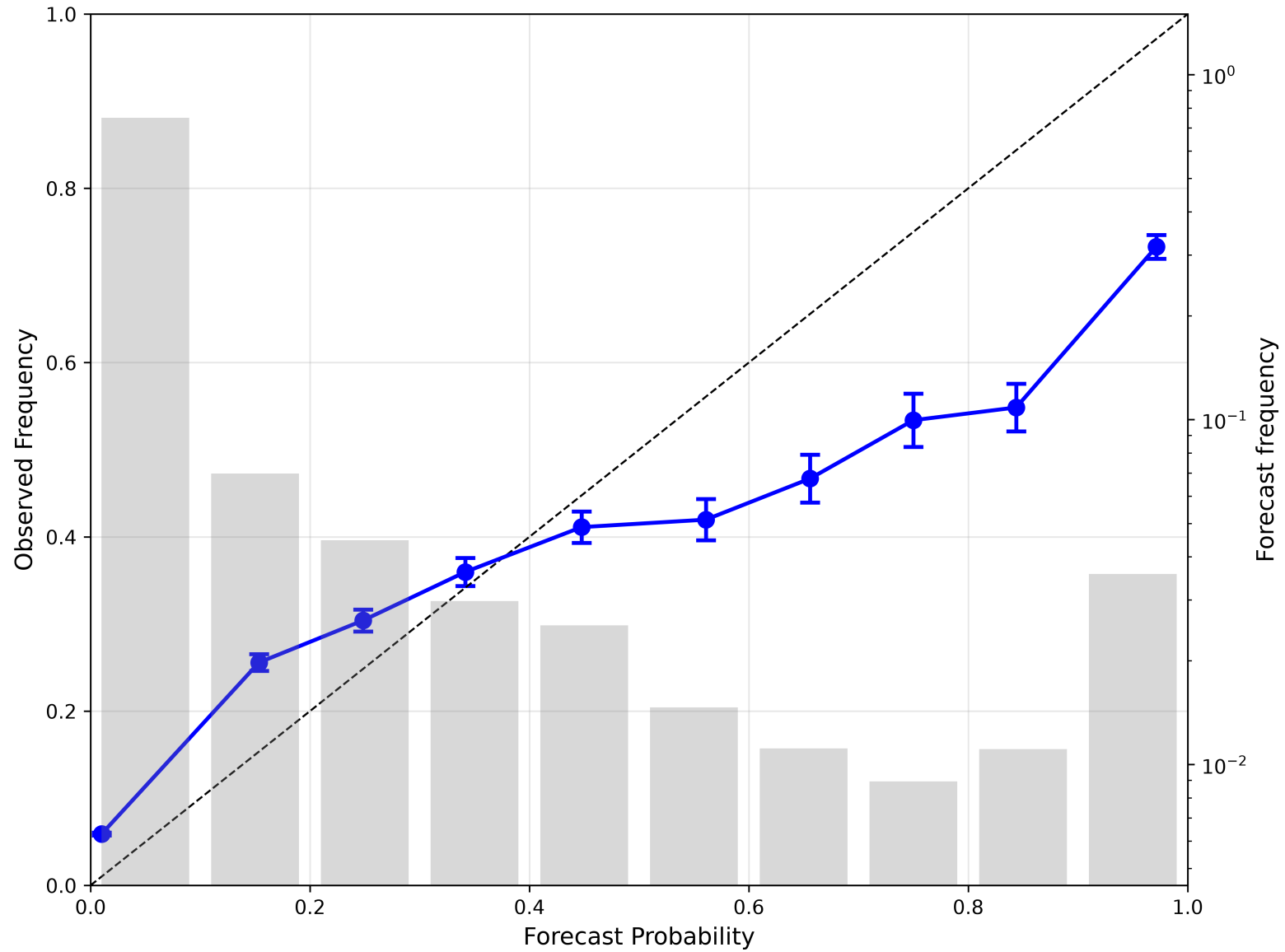
Reliability Diagram - GenCast - Days 16-30



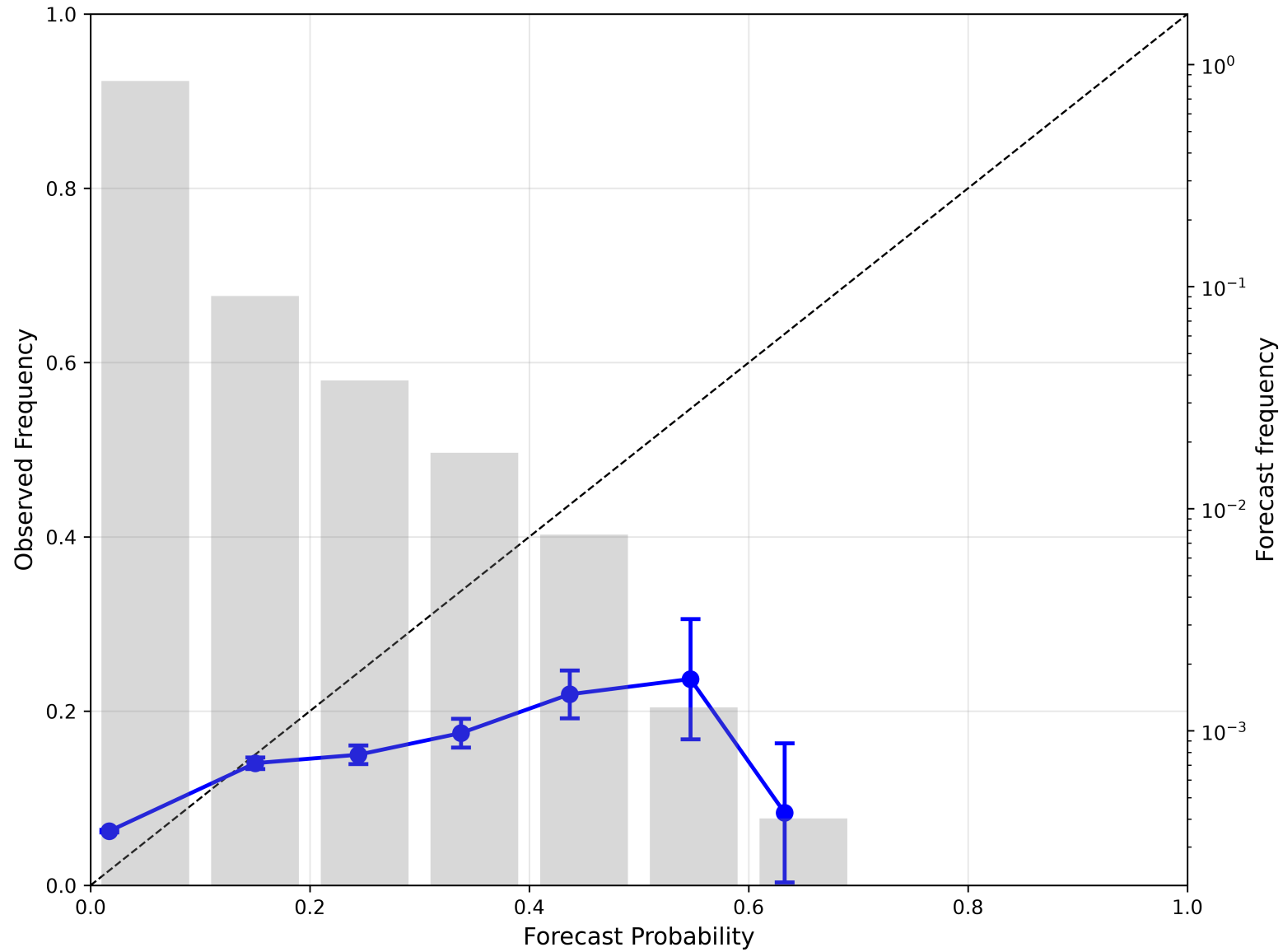
Reliability Diagram - AIFS Ensemble - Days 1-15



Reliability Diagram - AIFS Ensemble - Days 16-30

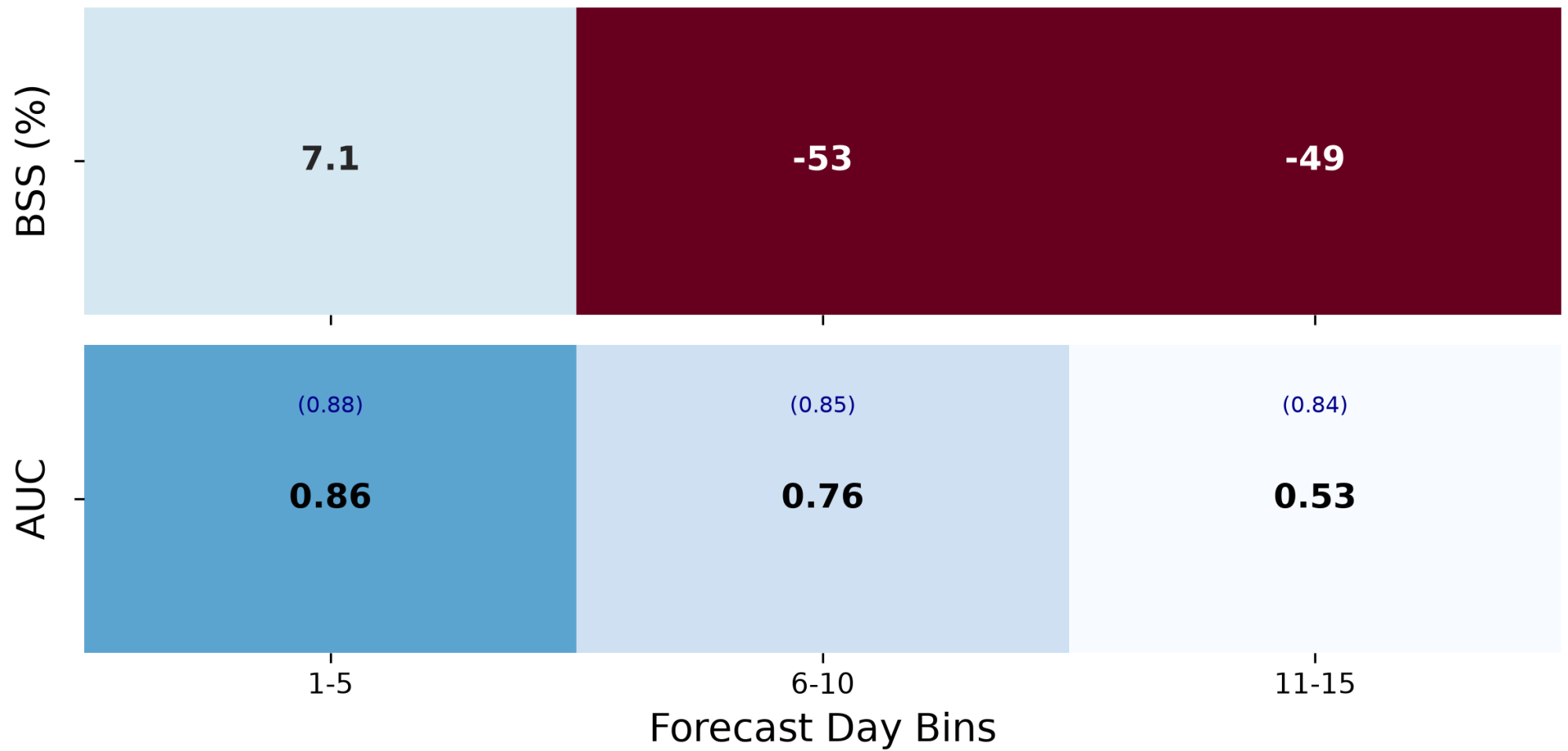


Reliability Diagram - GenCast - Days 1-15

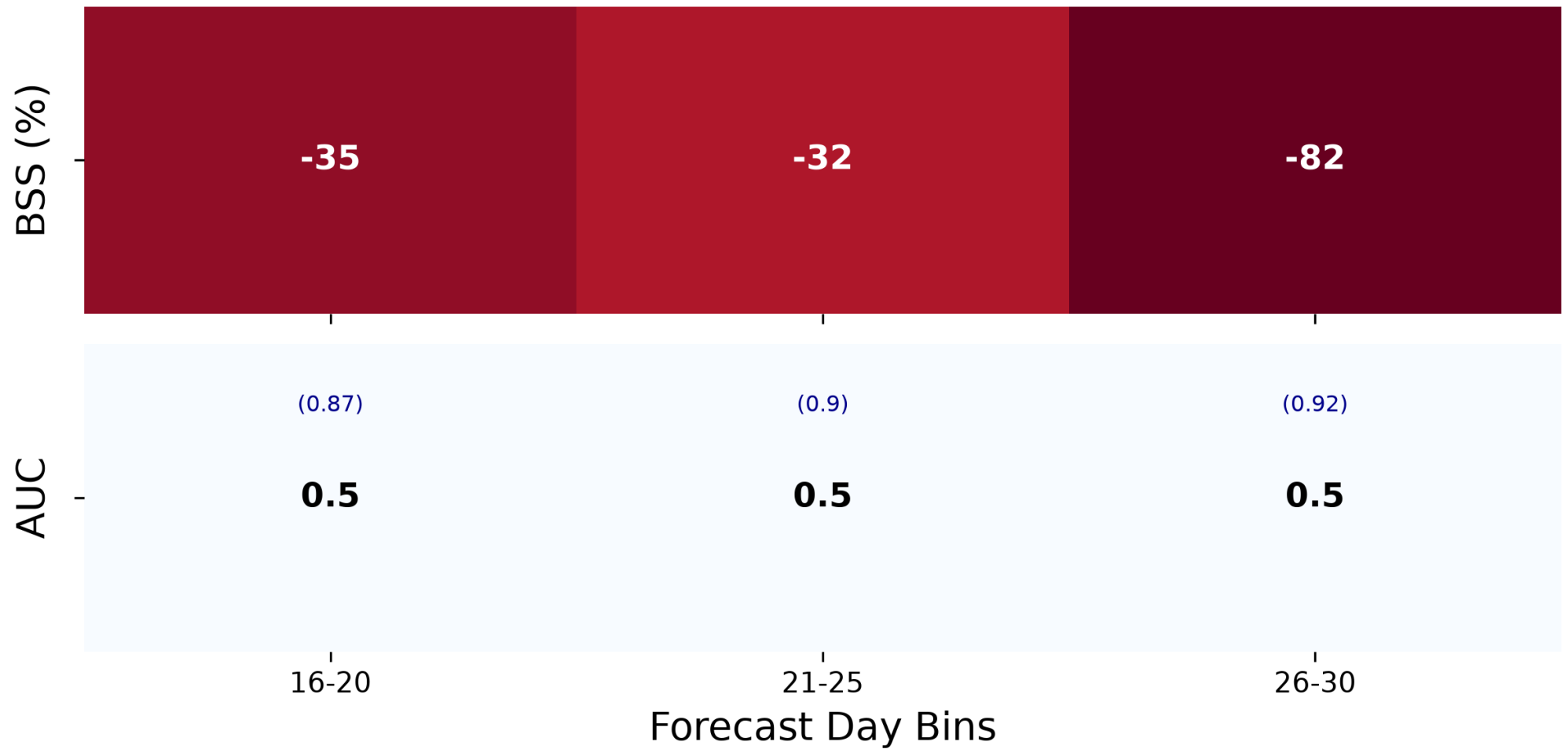


Reliability Diagram - GenCast - Days 16-30

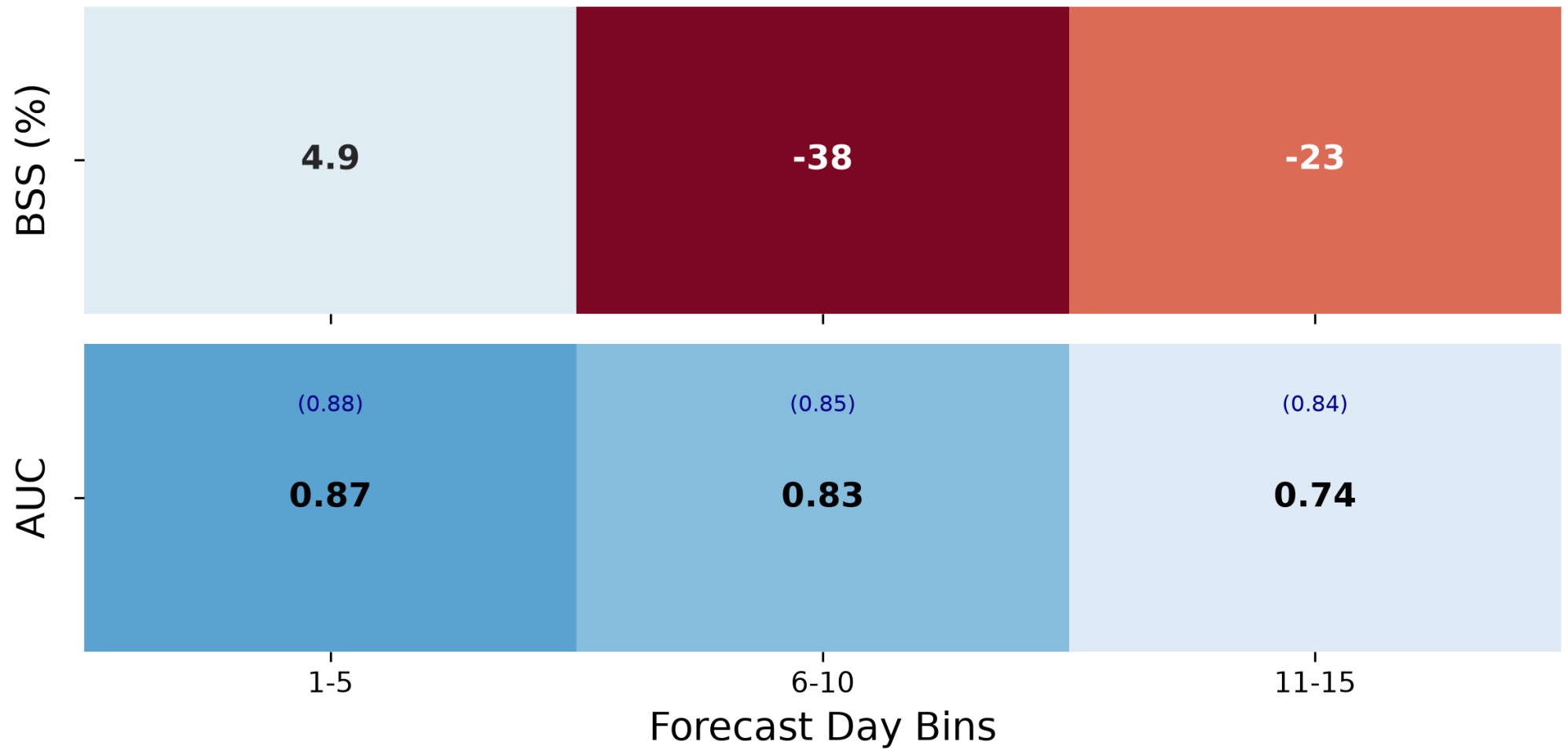
Skill Scores Heatmap (8)



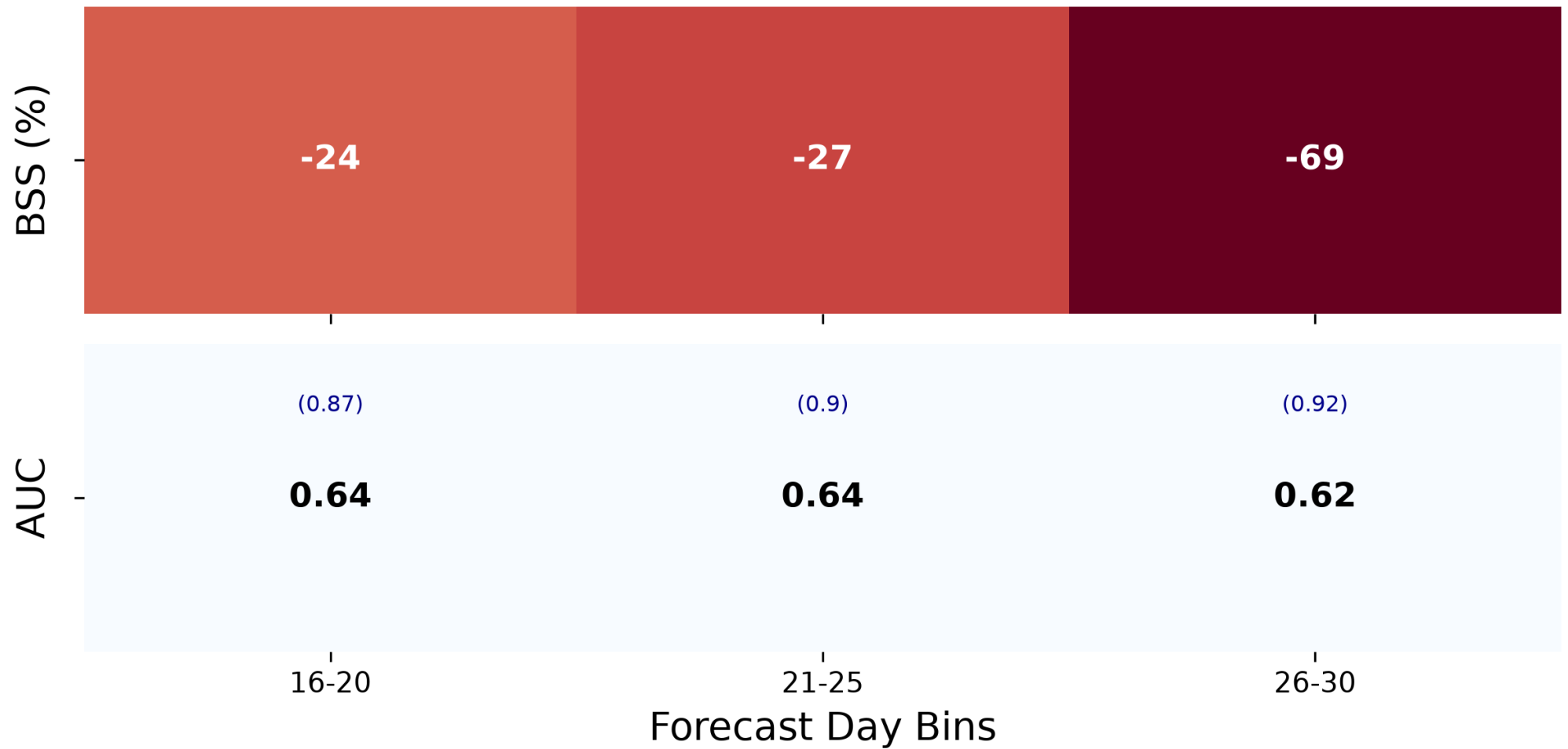
Skill Scores Heatmap - AIFS Ensemble - Days 1-15



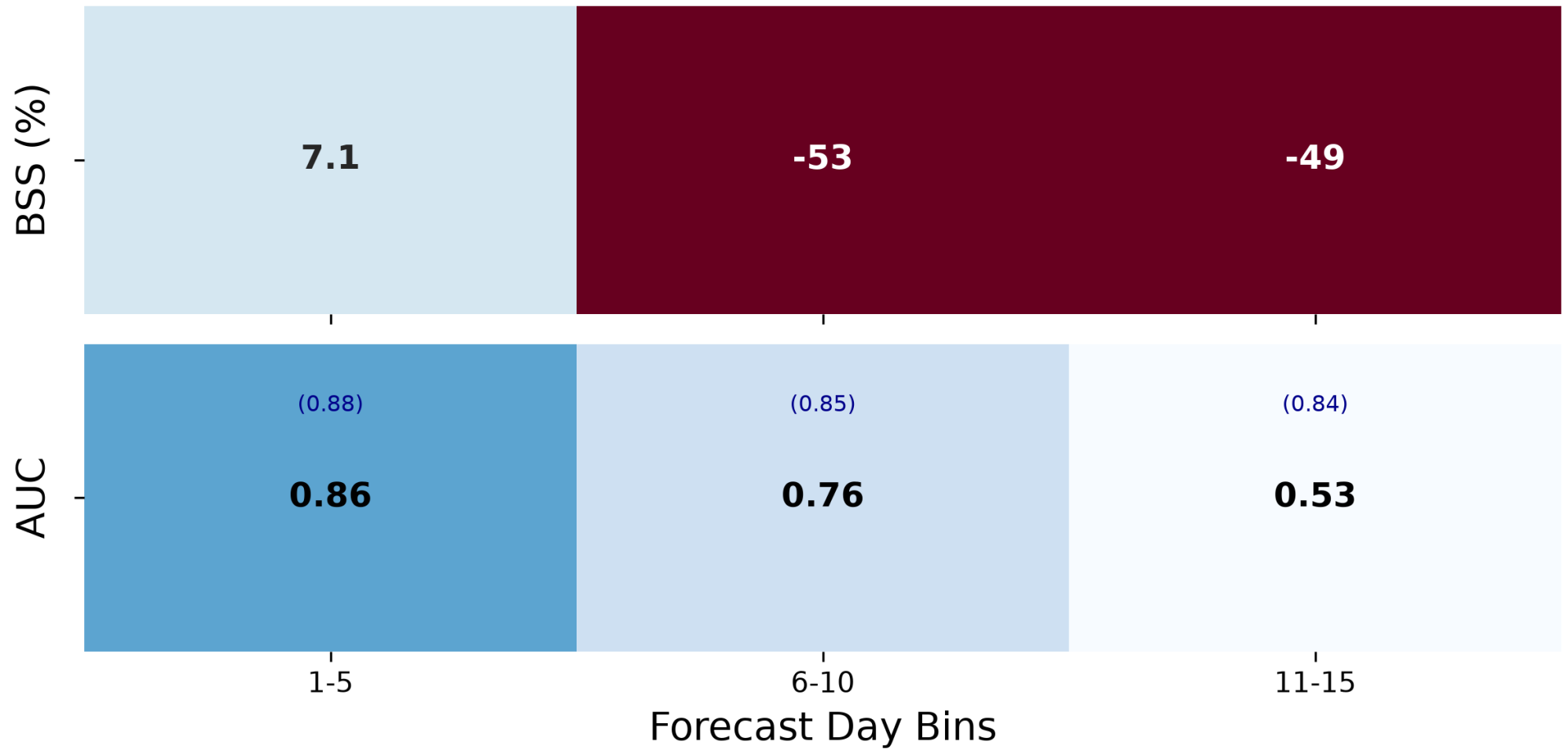
Skill Scores Heatmap - AIFS Ensemble - Days 16-30



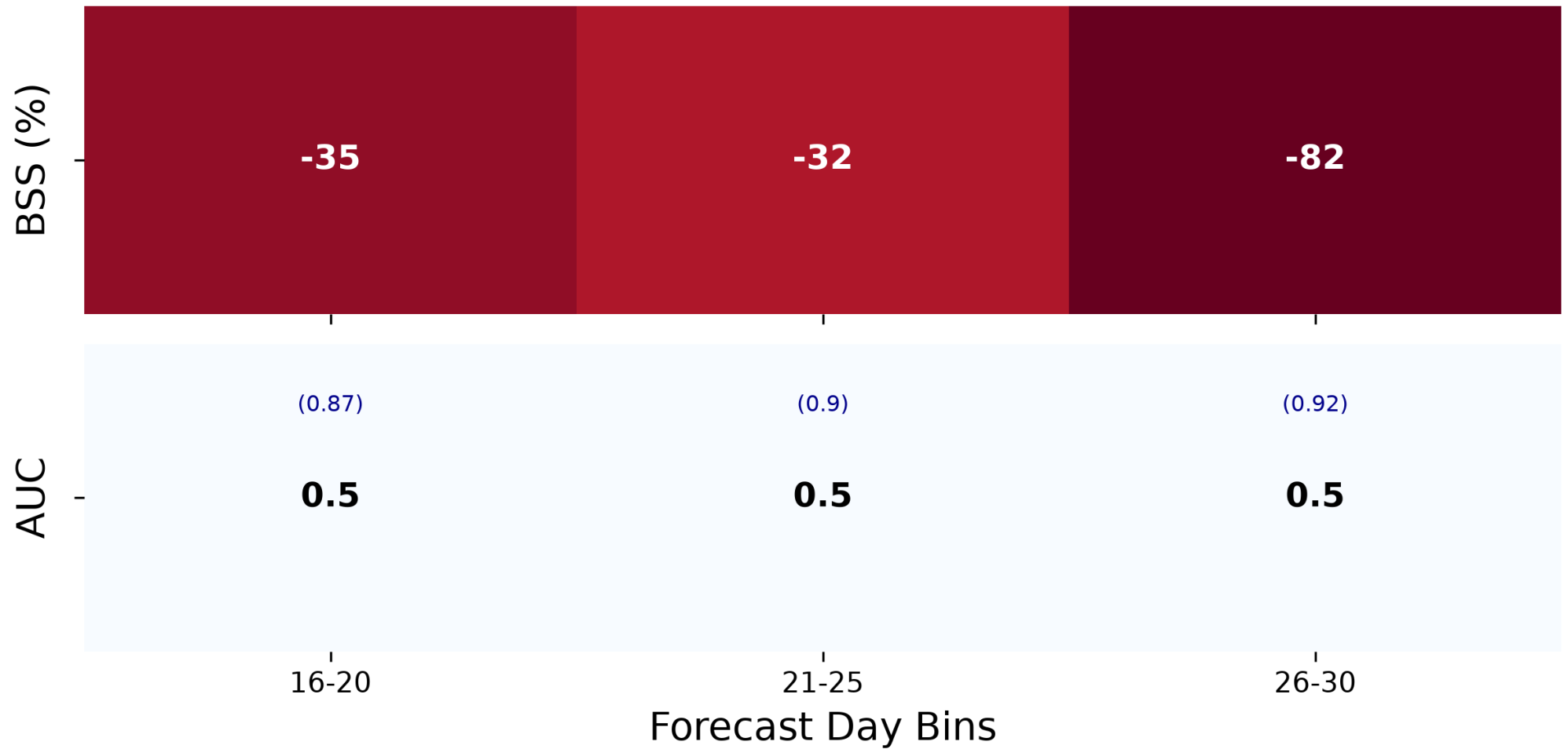
Skill Scores Heatmap - GenCast - Days 1-15



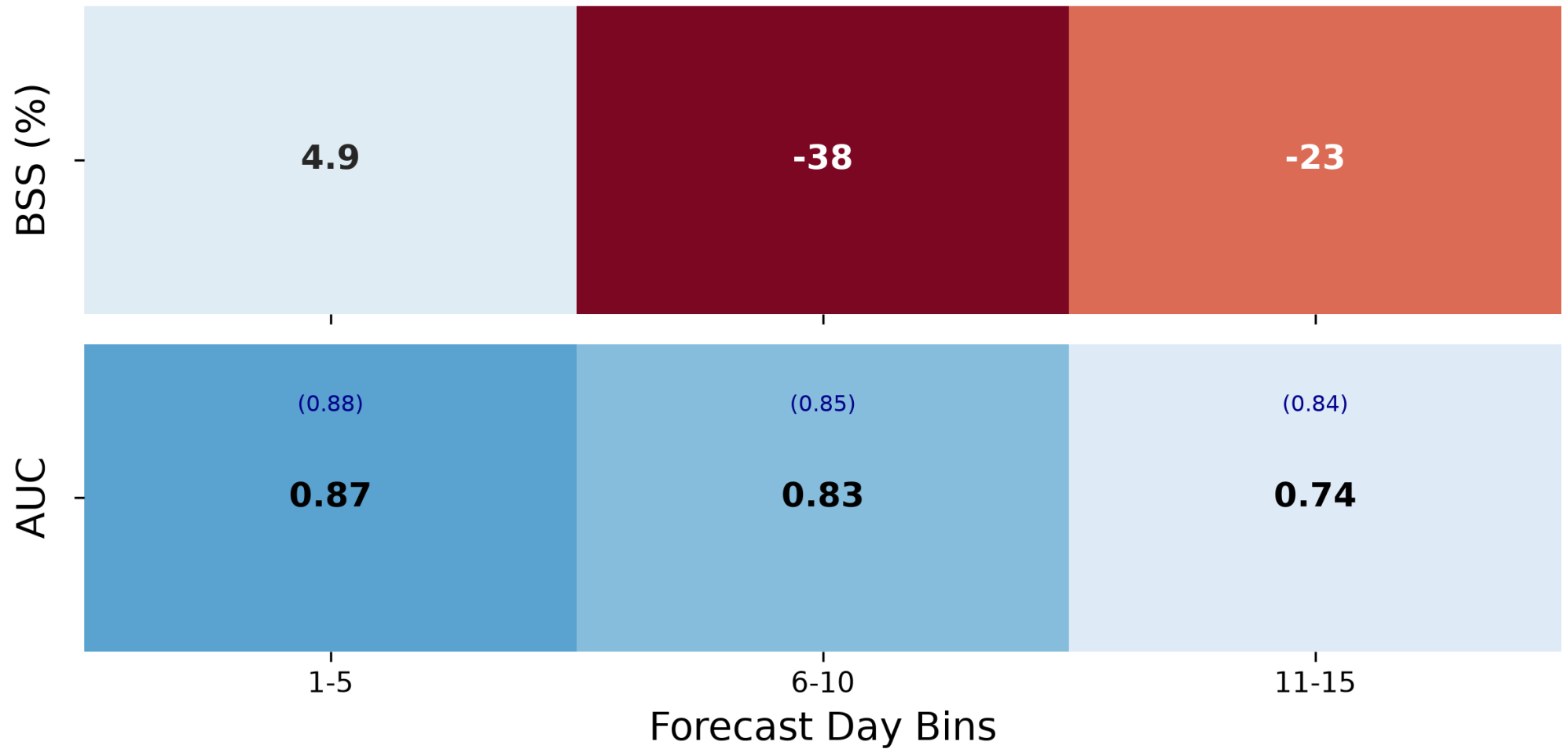
Skill Scores Heatmap - GenCast - Days 16-30



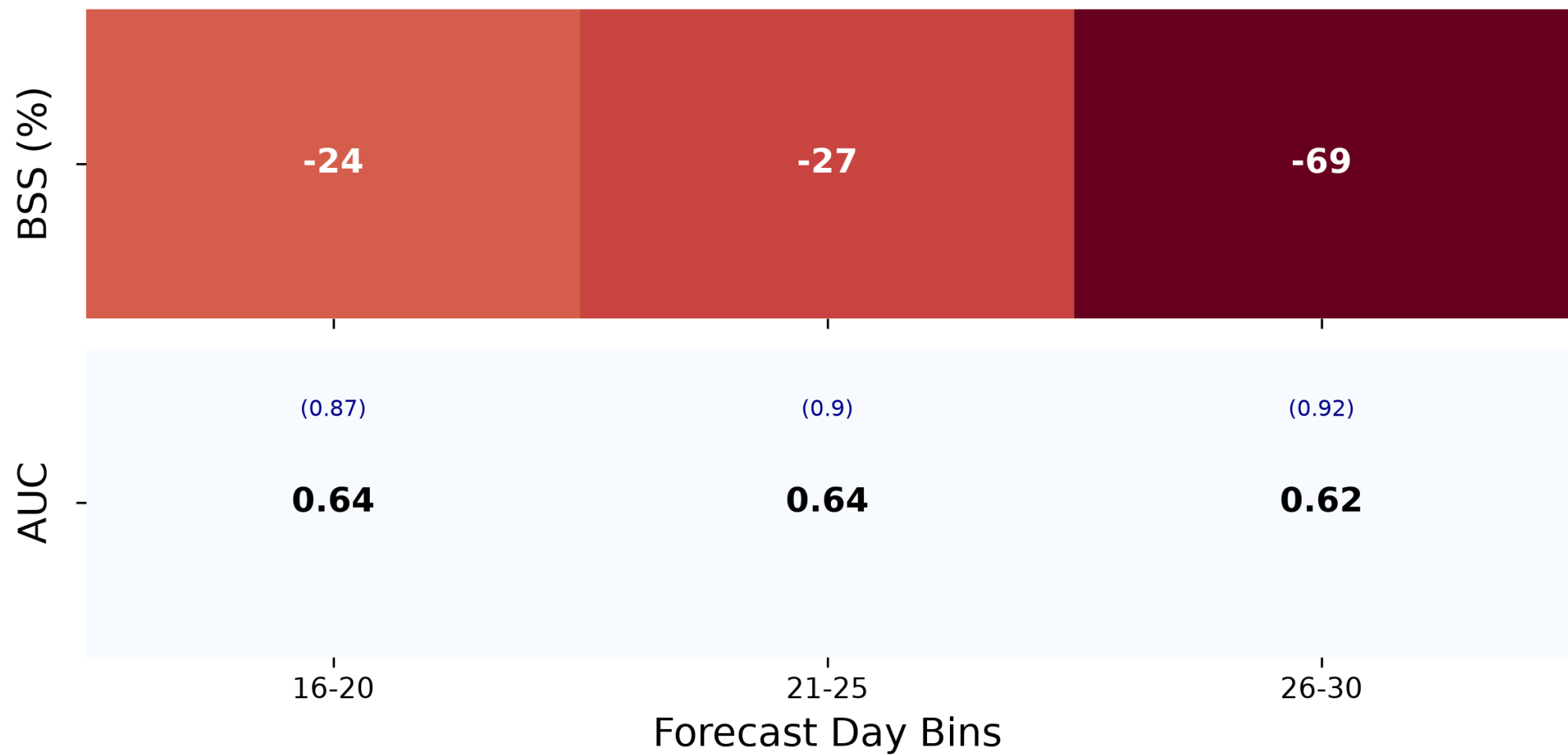
Skill Scores Heatmap - AIFS Ensemble - Days 1-15



Skill Scores Heatmap - AIFS Ensemble - Days 16-30



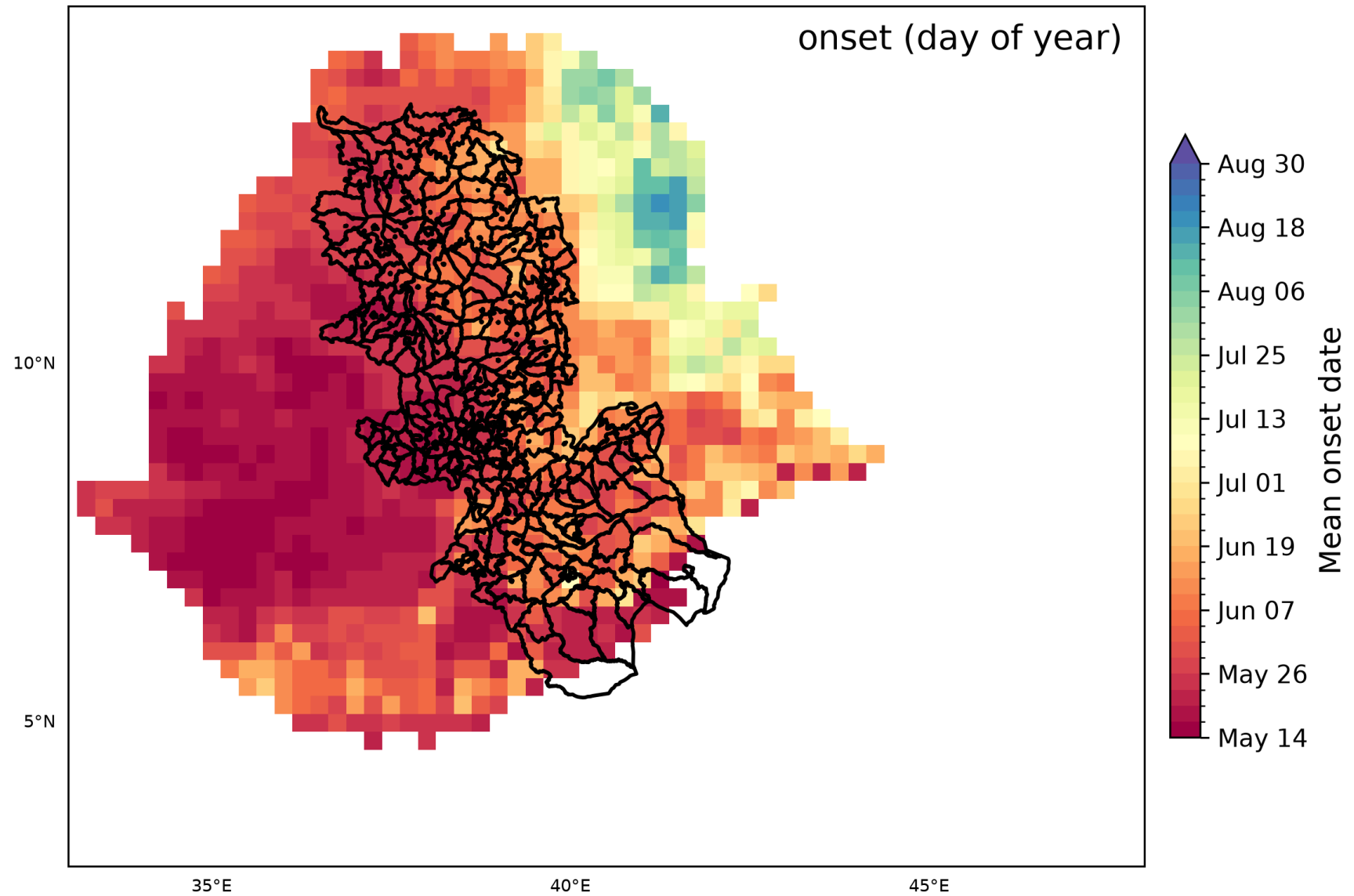
Skill Scores Heatmap - GenCast - Days 1-15



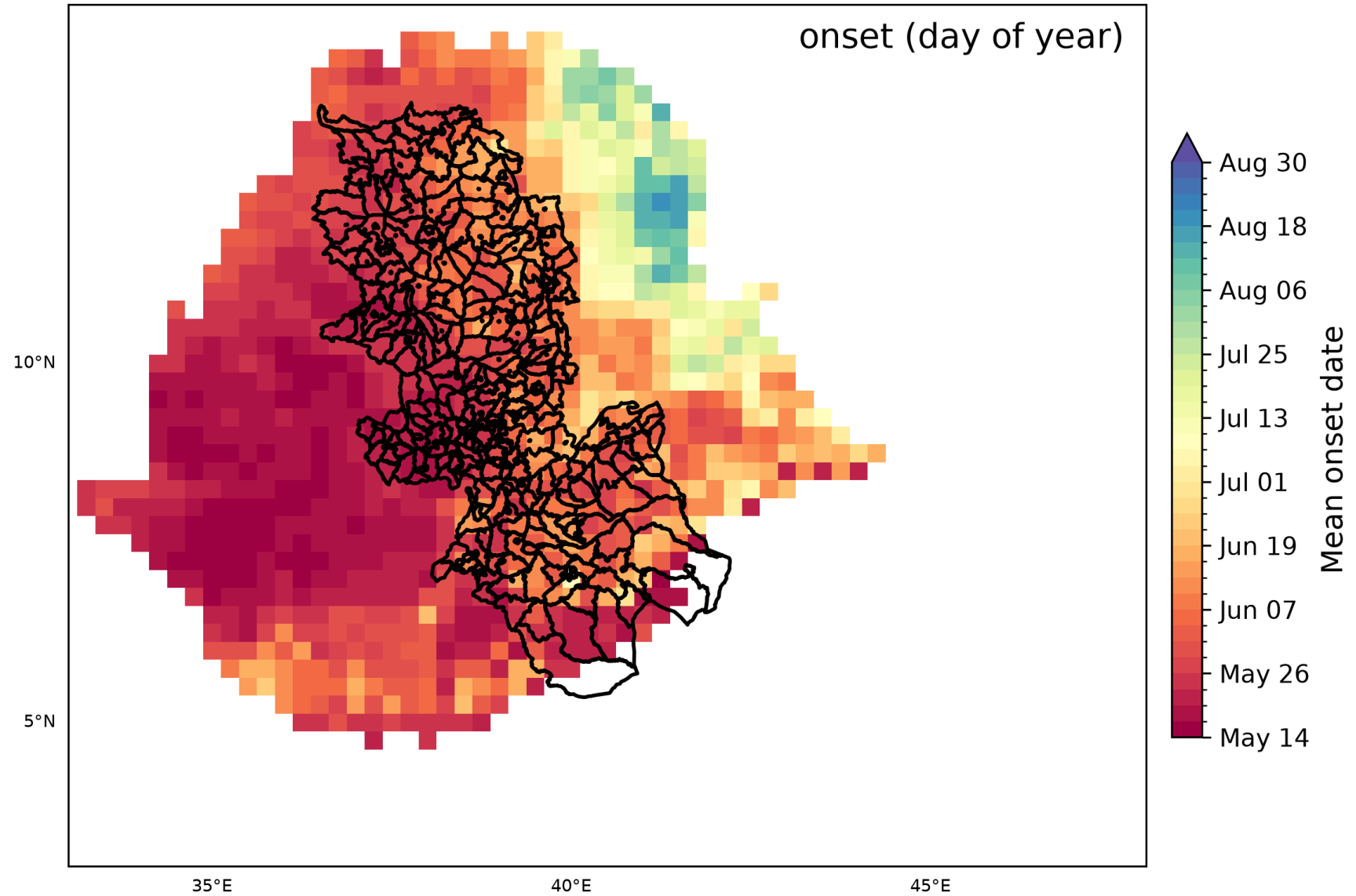
Skill Scores Heatmap - GenCast - Days 16-30

Climatology / Baseline

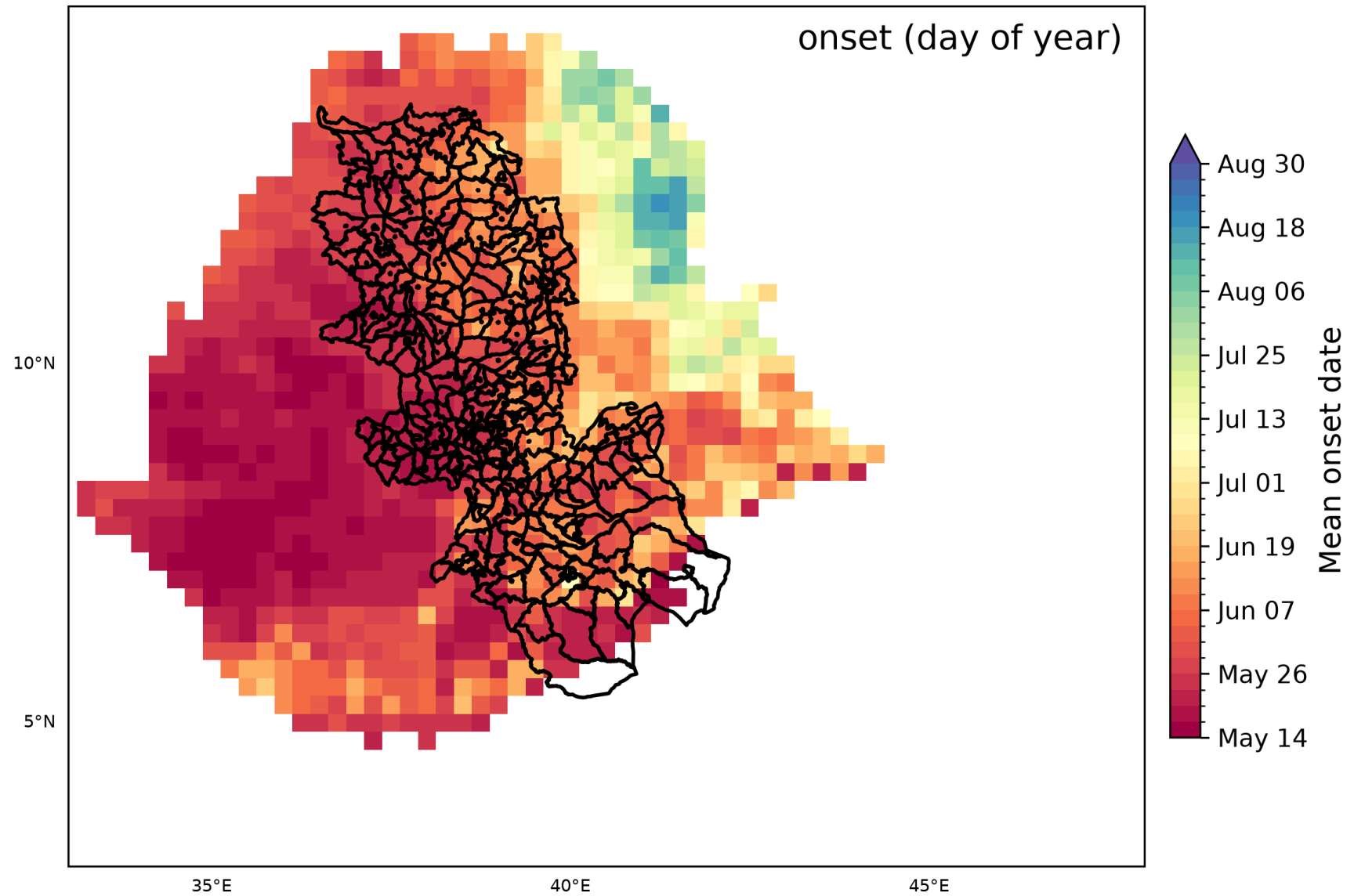
Climatology Onset (5)



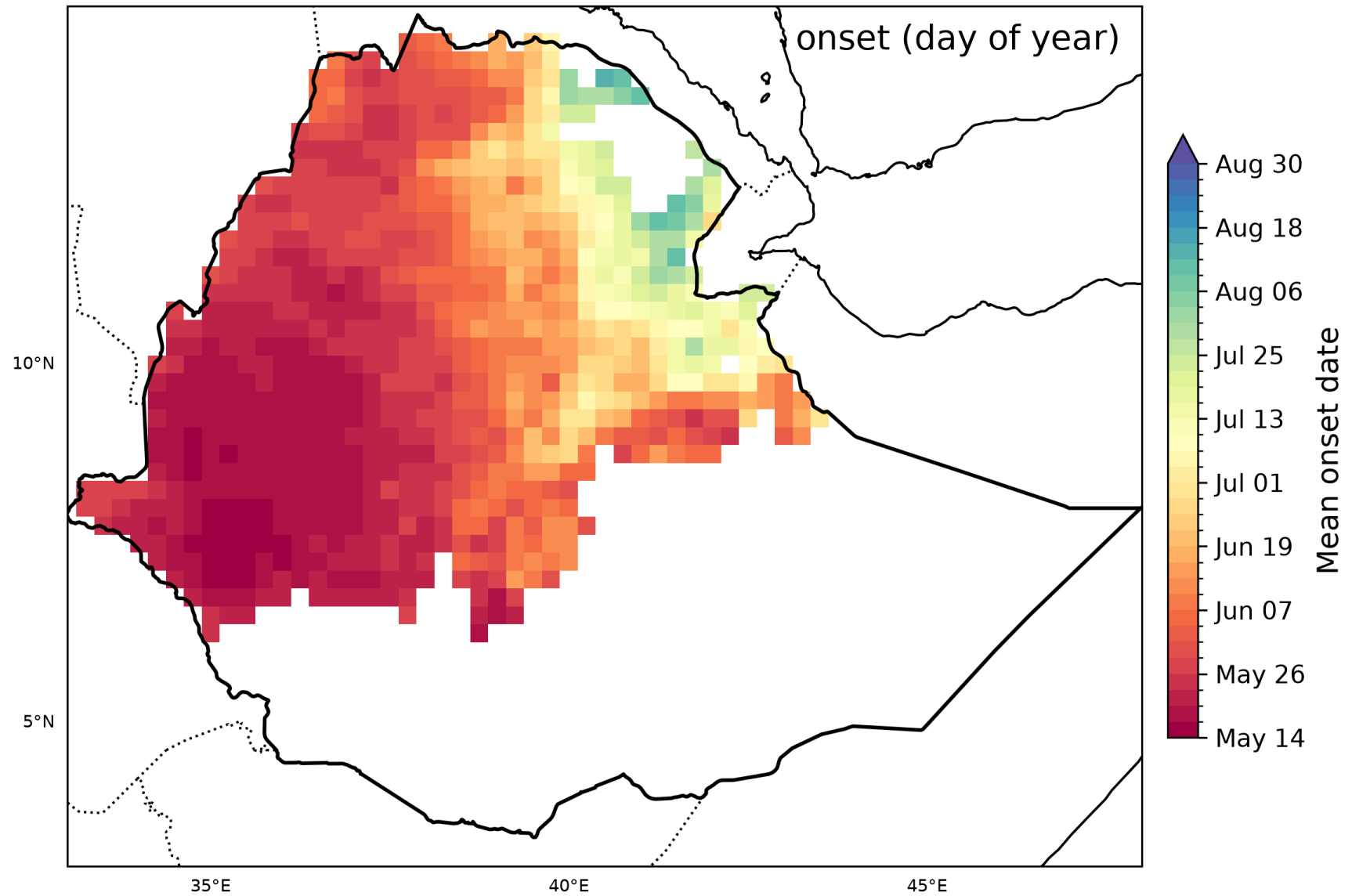
Climatology Onset - Climatology



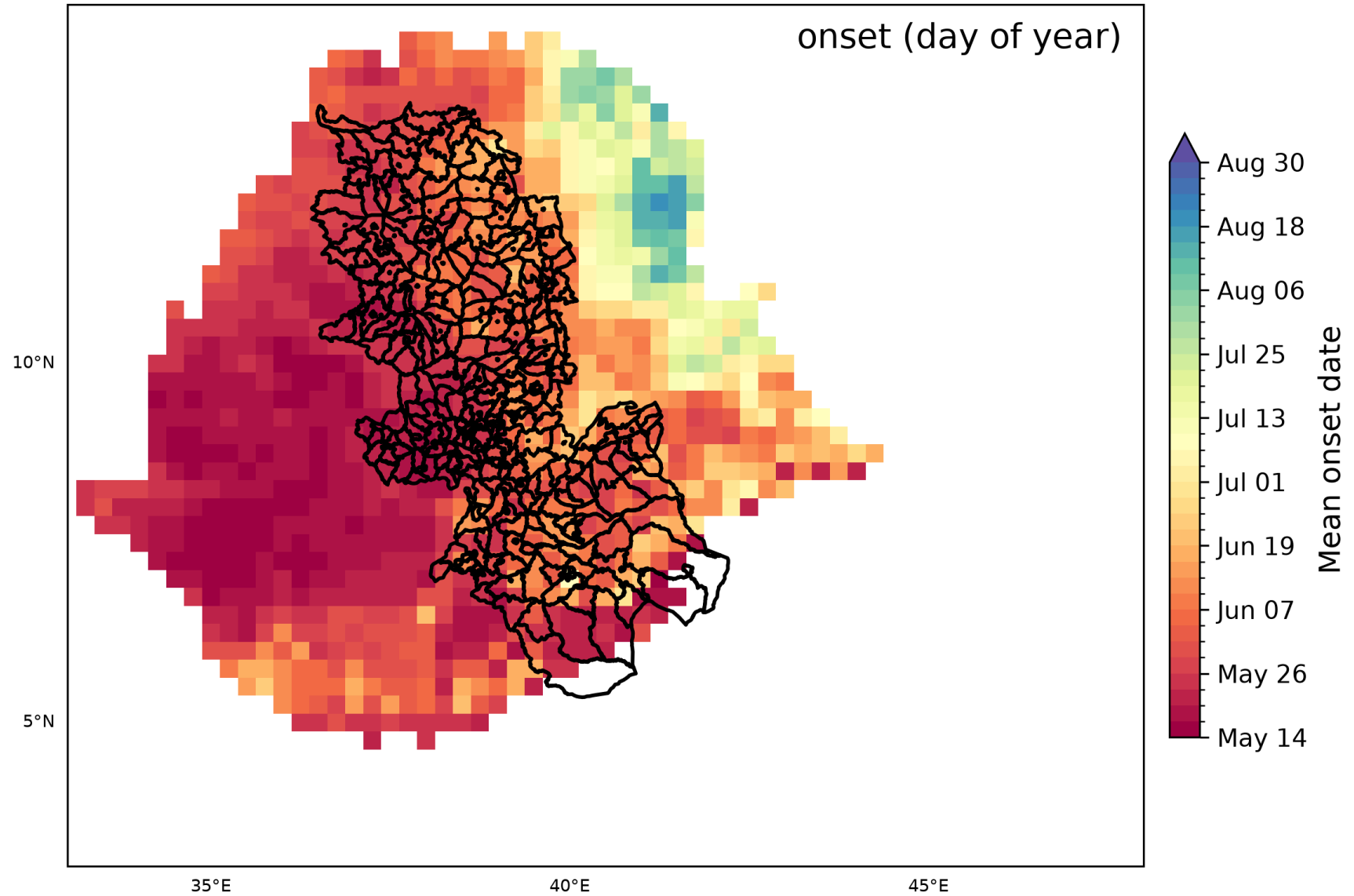
Climatology Onset - Climatology



Climatology Onset - Climatology



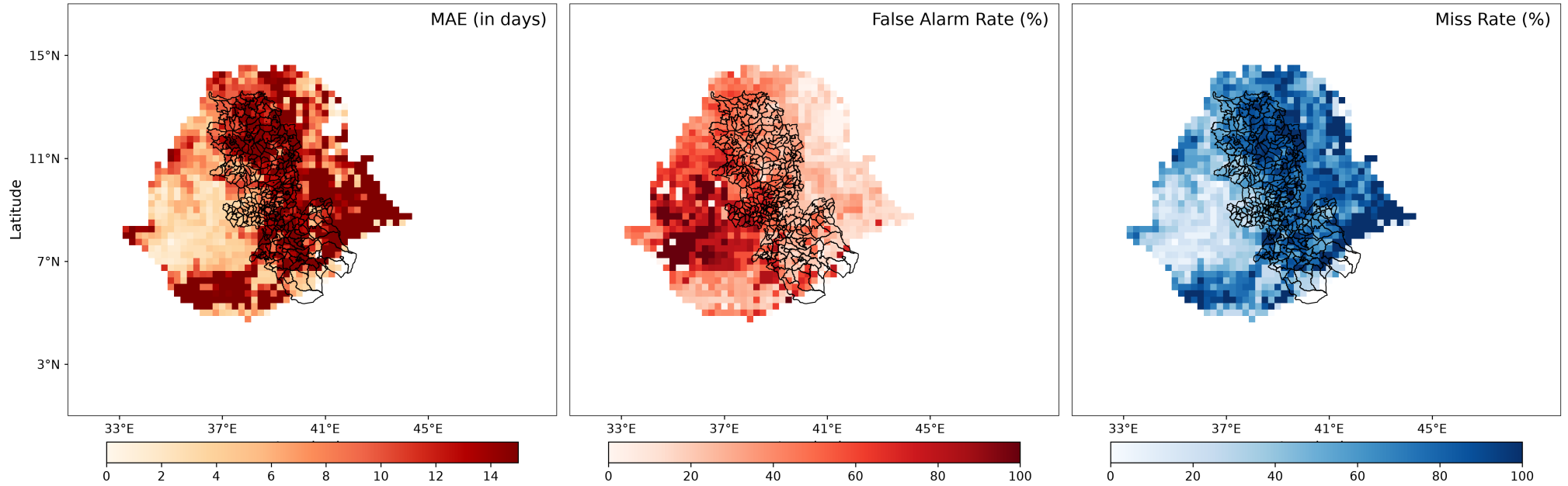
Climatology Onset - Climatology



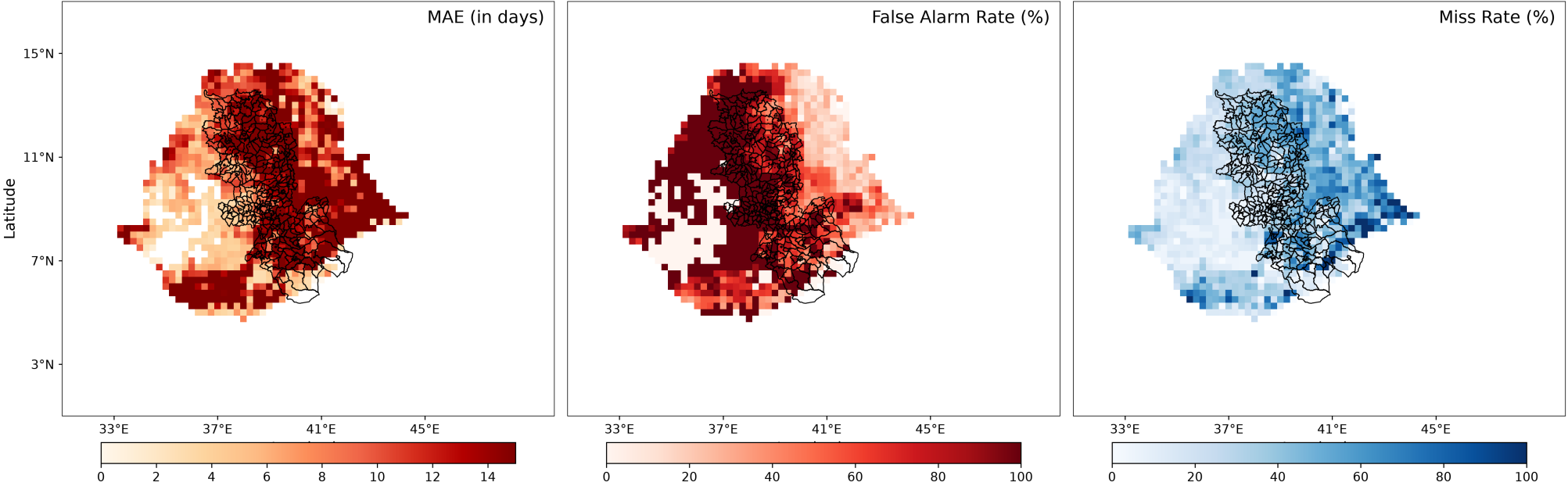
Climatology Onset - Climatology

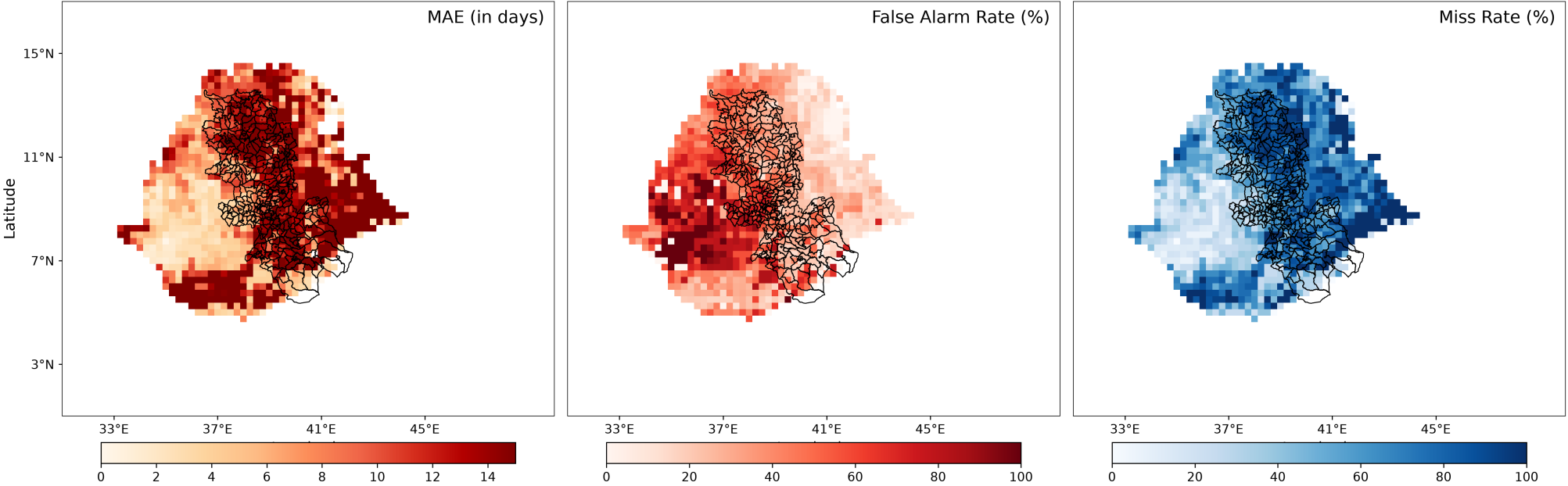
Forecast Benchmarking Report

Spatial Metrics (MAE / FAR / Miss Rate) (8)

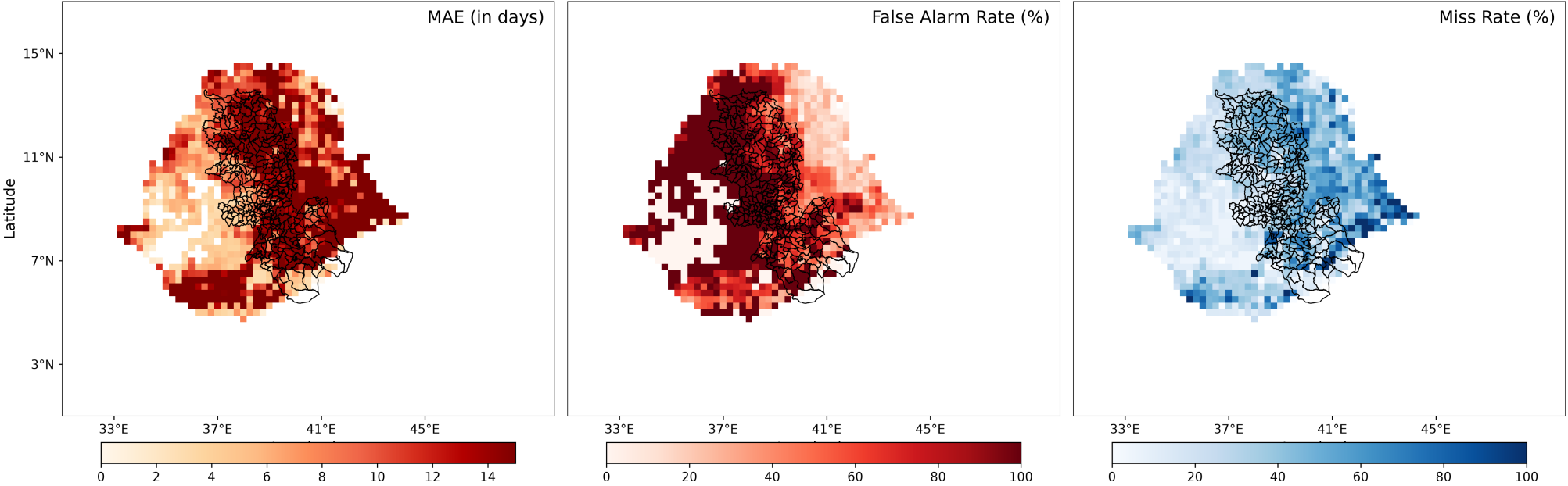


Spatial Metrics: Climatology (Days 1-15)

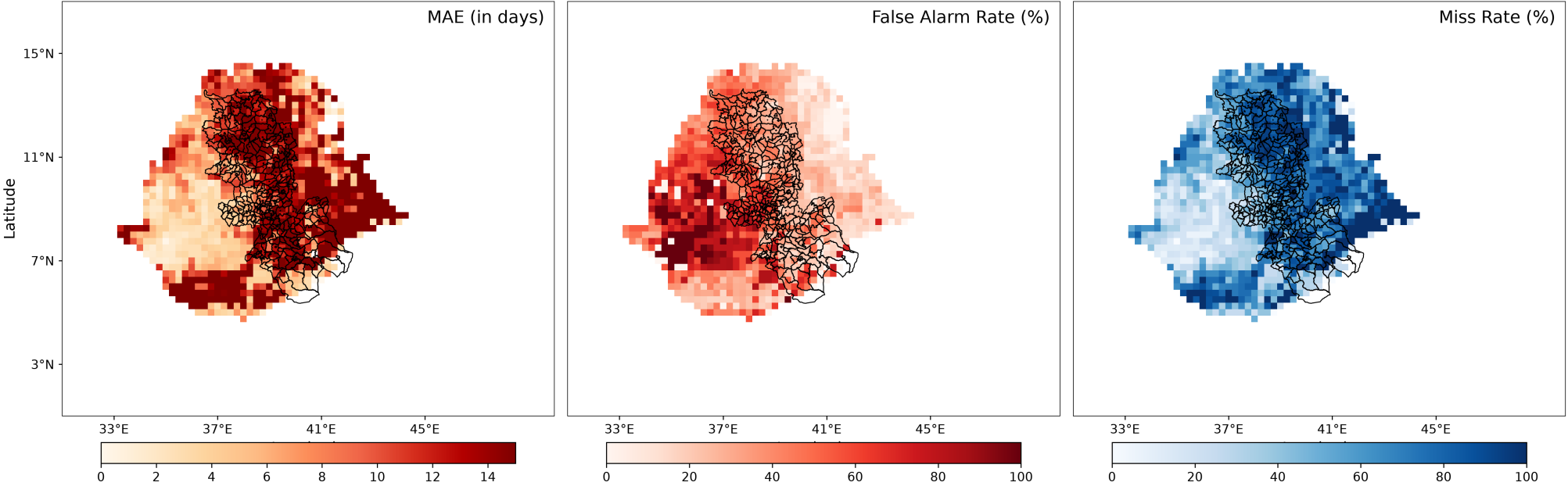




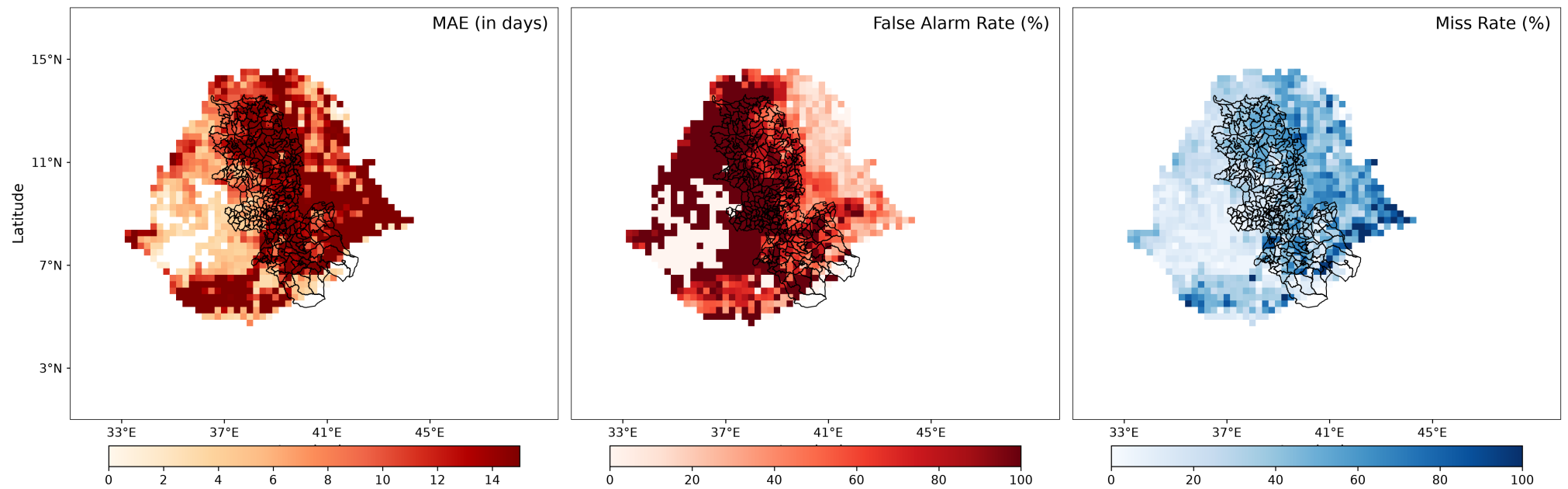
Spatial Metrics: Climatology (Days 1-15)



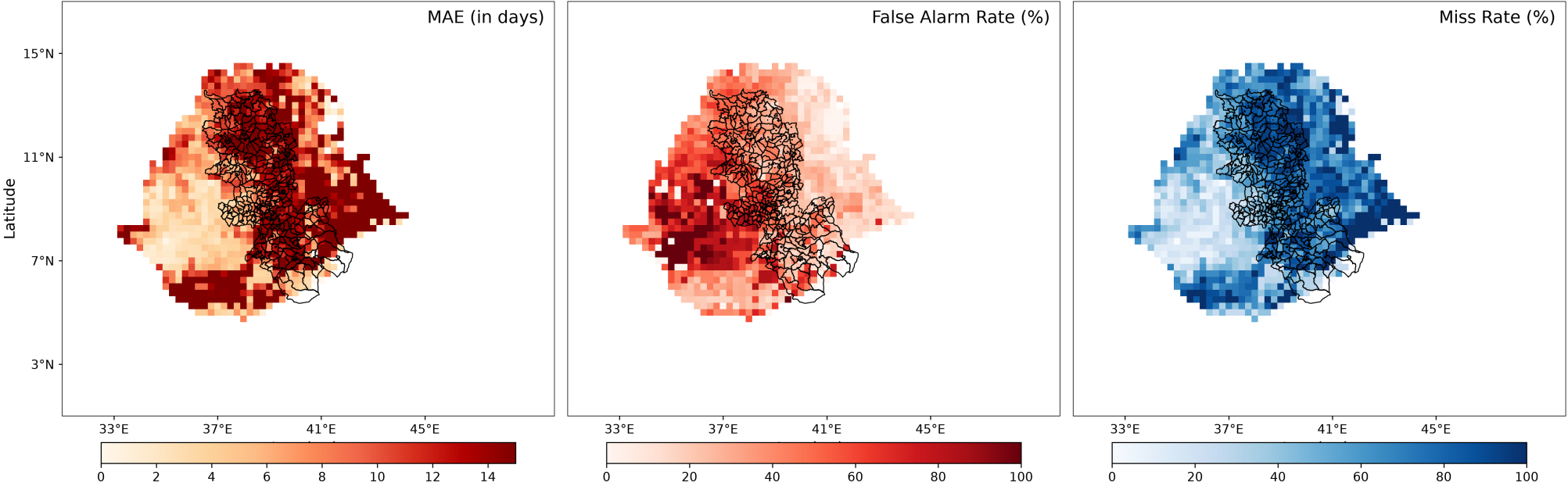
Spatial Metrics: Climatology (Days 16-30)



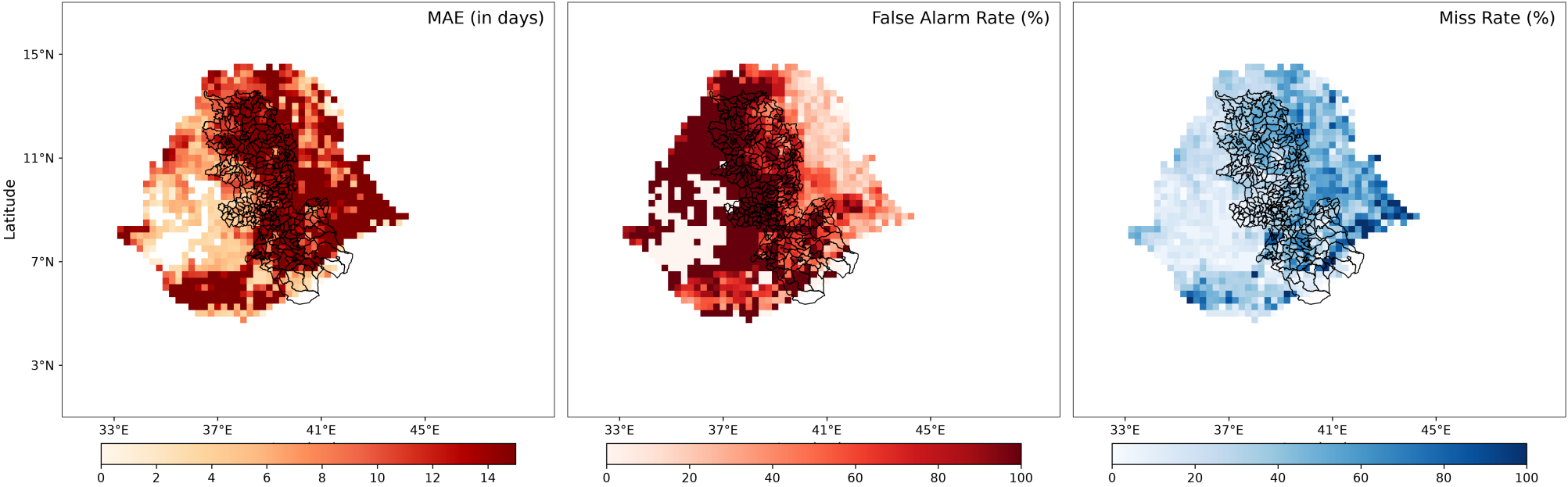
Spatial Metrics: Climatology (Days 1-15)



Spatial Metrics: Climatology (Days 16-30)



Spatial Metrics: Climatology (Days 1-15)

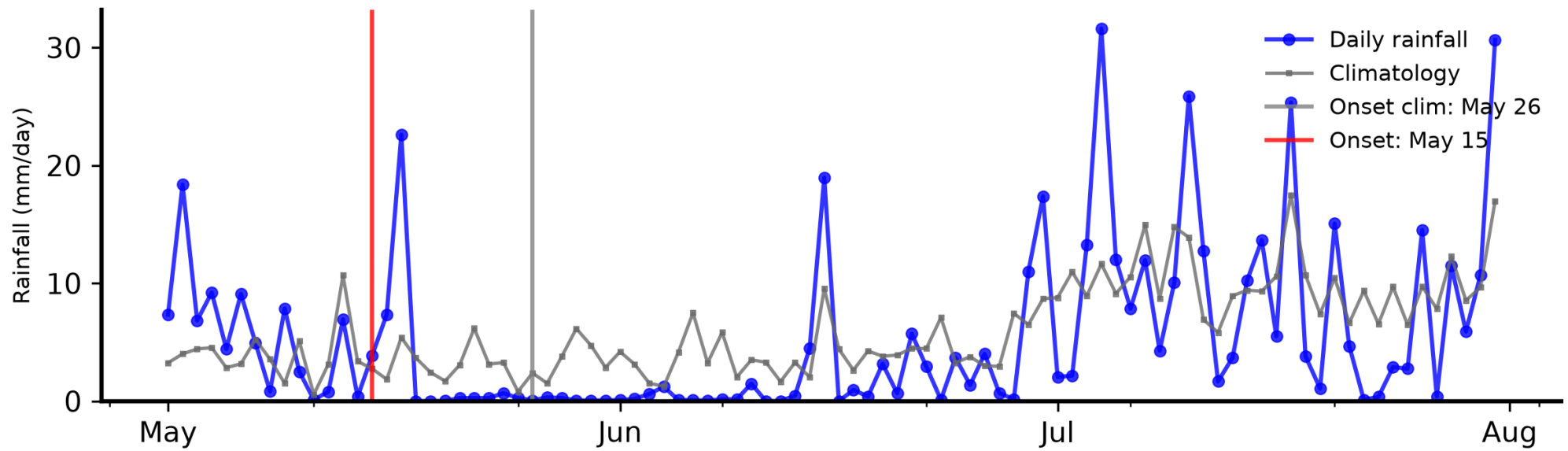


Spatial Metrics: Climatology (Days 16-30)

Other

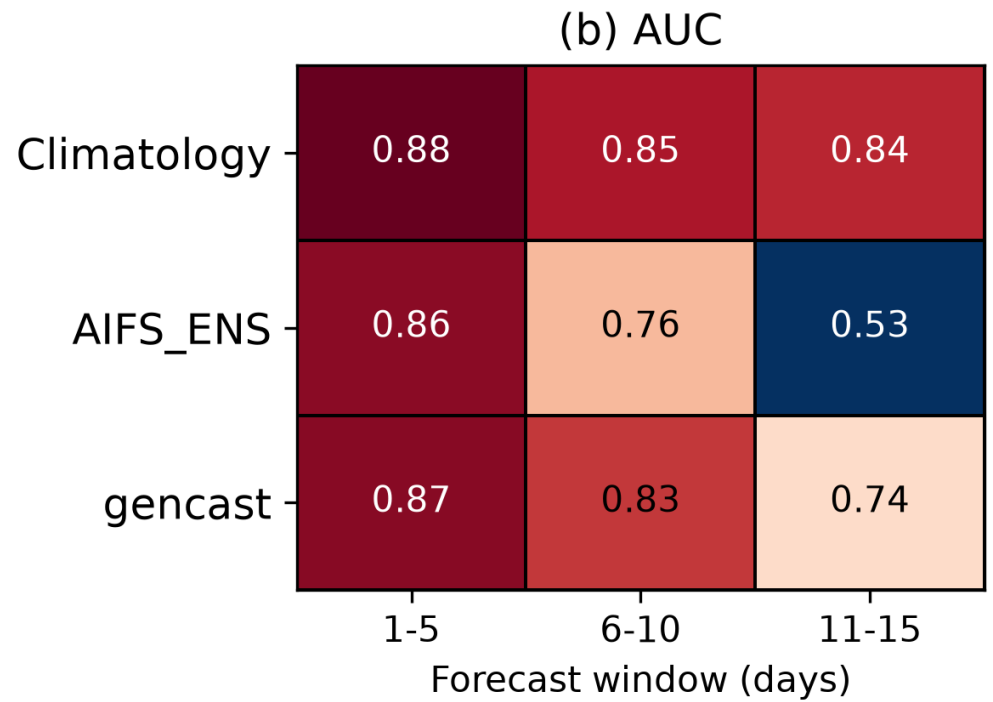
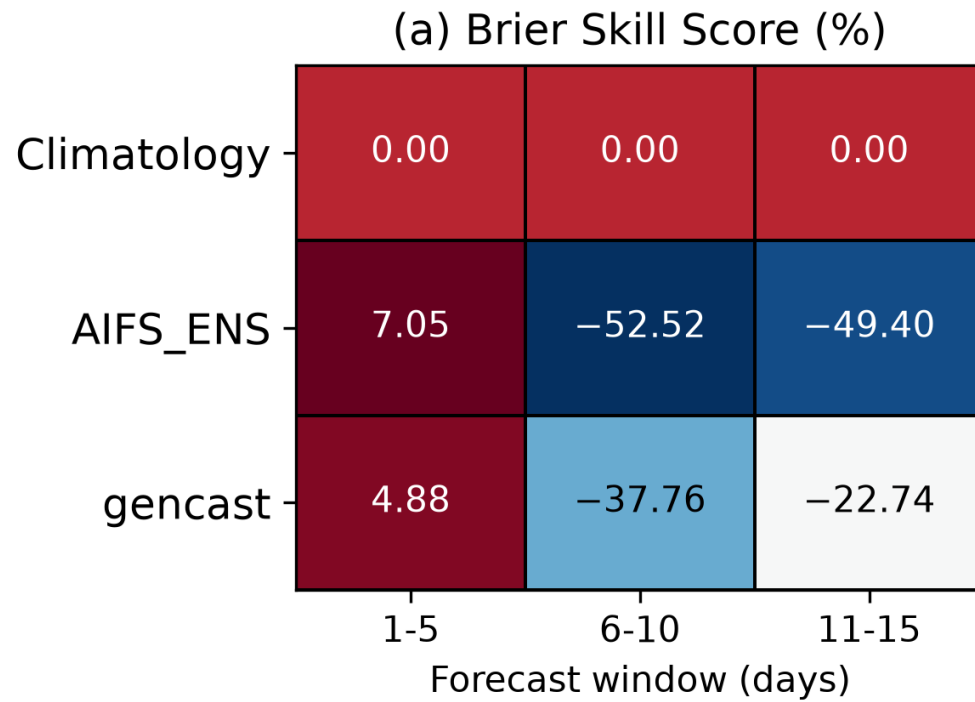
Other (1)

Daily Rainfall Onset Dates - 2021
Location: 9.5°N, 38.5°E

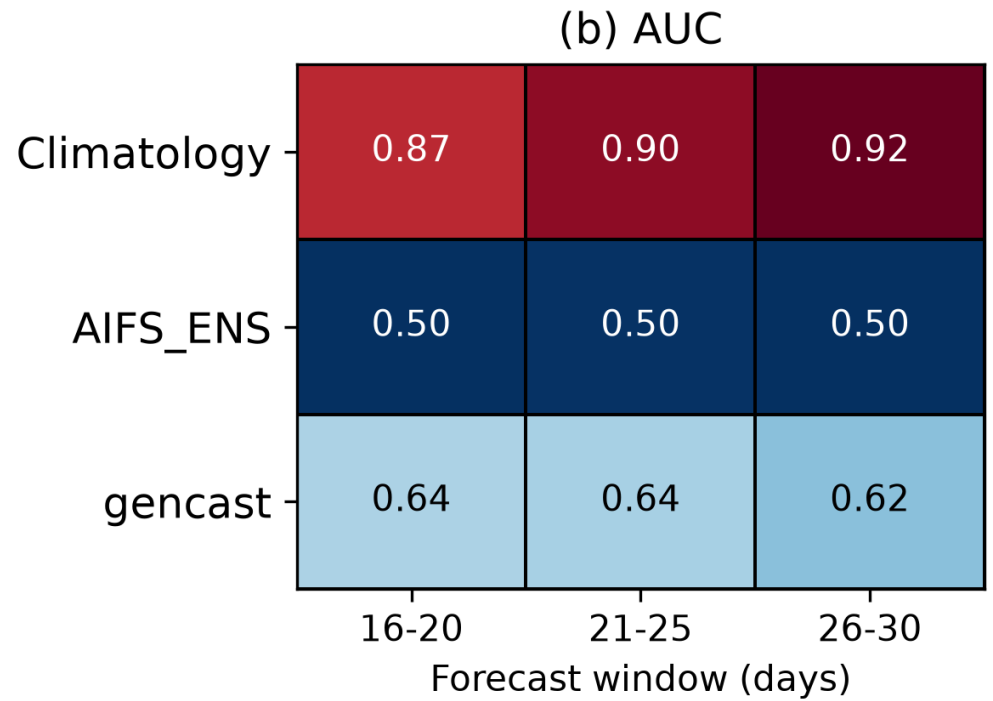
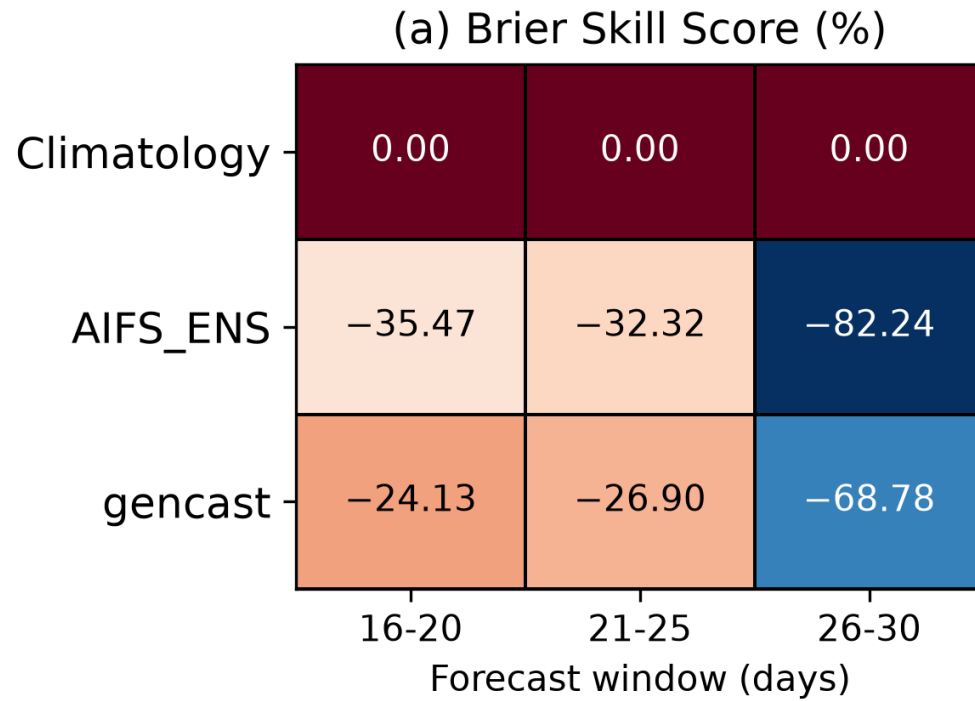


Onset Time Series Lat9.5 Lon38.5 2021

Portrait Panel (2)



Panel Portrait - Days 1-15



Panel Portrait - Days 16-30

Notes

This report was generated automatically from the benchmark dashboard. Metrics are computed from model output files available at generation time.